

Lecture 8: Introduction to Data Visualization

Heather Mattie, PhD

September 24, 2026

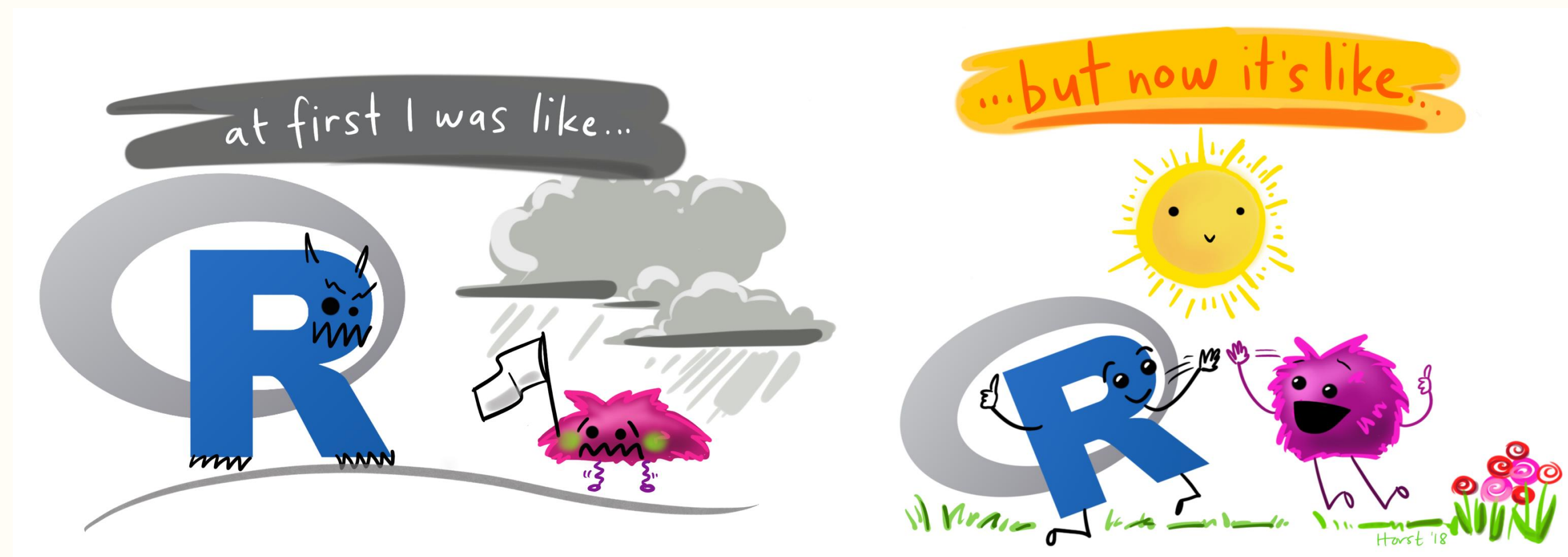
Recipe of the Day!

Authentic massaman curry



Announcements

- Heather's office hour: Thursday (today) 12-1pm
- Homework #2 due Friday, October 9th by 11:59pm
- Lab will be held this week



Coding question of the day

Using the **gapminder** dataset from **ds1labs**, find each country's life expectancy in 1960 and in 2010, and compute how much it changed. Hint: it may be useful to create a data frame with rows where year is equal to 1960 and a separate data frame with rows where year is equal to 2010 and then join them before calculating the difference.

Be sure to run this code first

```
library(ds1labs)
library(tidyverse)

data(gapminder)
```

Introduction to Data Visualization

Data Visualization Roadmap

1. Introduction to `ggplot2`
2. Fine-tuning visualizations
3. Data visualization principles
4. Case study: vaccines

Introduction to data visualization

Introduction to **ggplot2**

Heather Mattie, PhD

The Data Science Pipeline



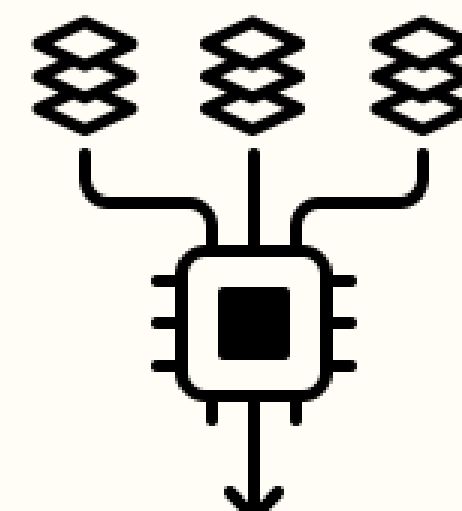
**Collecting
and
importing
(loading)
the data**



**Processing
(cleaning,
wrangling)
the data**



**Visualizing
and
summarizing
the data**



**Building
models
(statistical
and ML)**



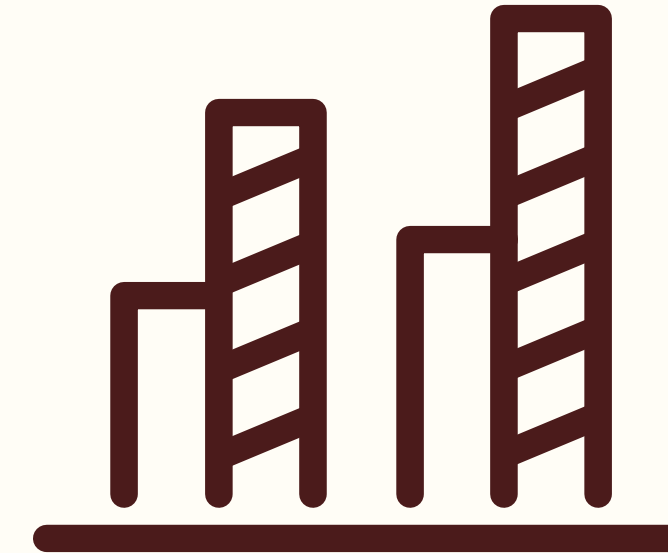
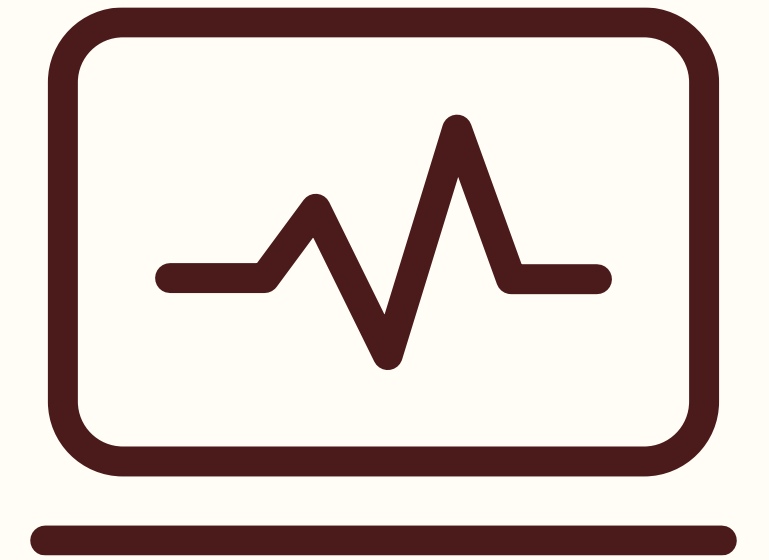
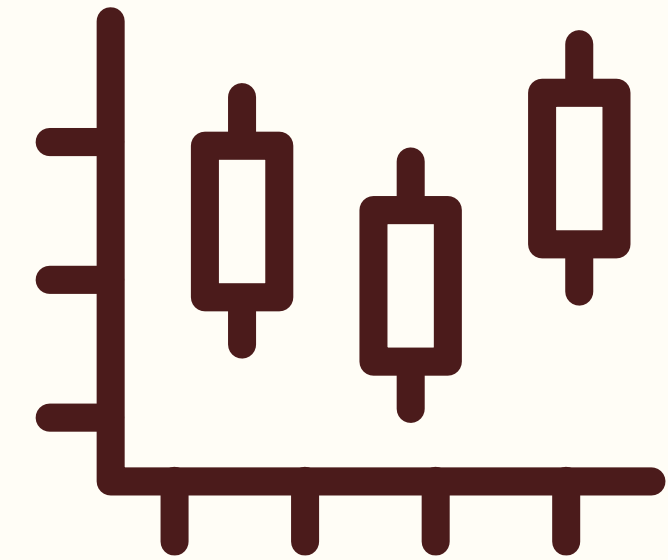
**Interpretation
and
communication
of results**

Data visualization

Data visualization is a powerful way to **communicate** data-driven findings

Data visualization is also used for **exploratory data analysis (EDA)**

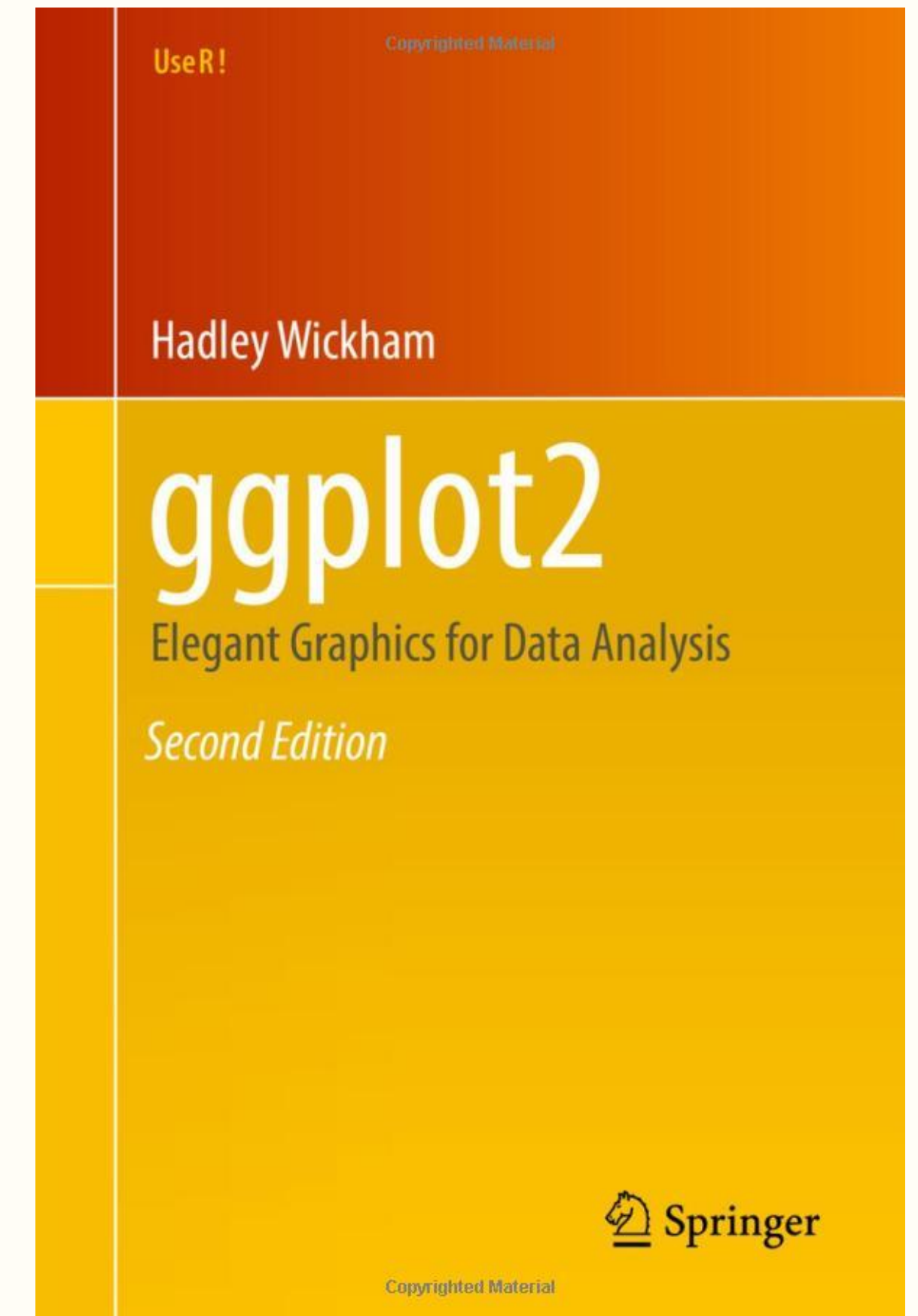
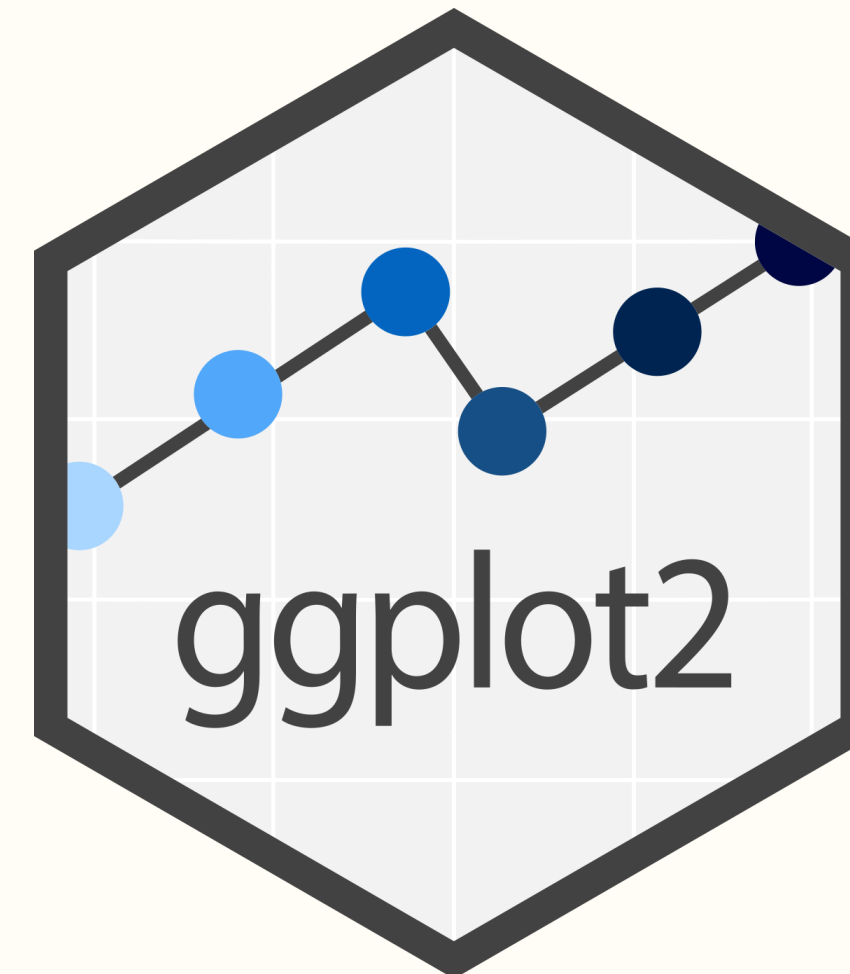
- View data distributions
- Detect outliers
- Show patterns in the data



Data visualization with `ggplot2`

`ggplot2` is a data visualization package

- Structured way to create a wide range of high-quality, impactful data visualizations
- Intuitive building blocks
- Concise code



Data visualization with `ggplot2`

The `gg` in `ggplot2` refers to *grammar of graphics*

- By learning a handful of ggplot2 building blocks and grammar, hundreds of different plots can be created
- **Data** – the data used for the plot
- **Geometry** – the type of plot created
- **Aesthetic mapping** – how the data is mapped to the plot

Anatomy of a `ggplot2` visualization

The code for a typical `ggplot2` visualization will look like this:

```
data %>%
```

```
  ggplot() +
```

```
  Layer 1 +
```

```
  Layer 2 +
```

```
  Layer 3 +
```

```
  ...
```

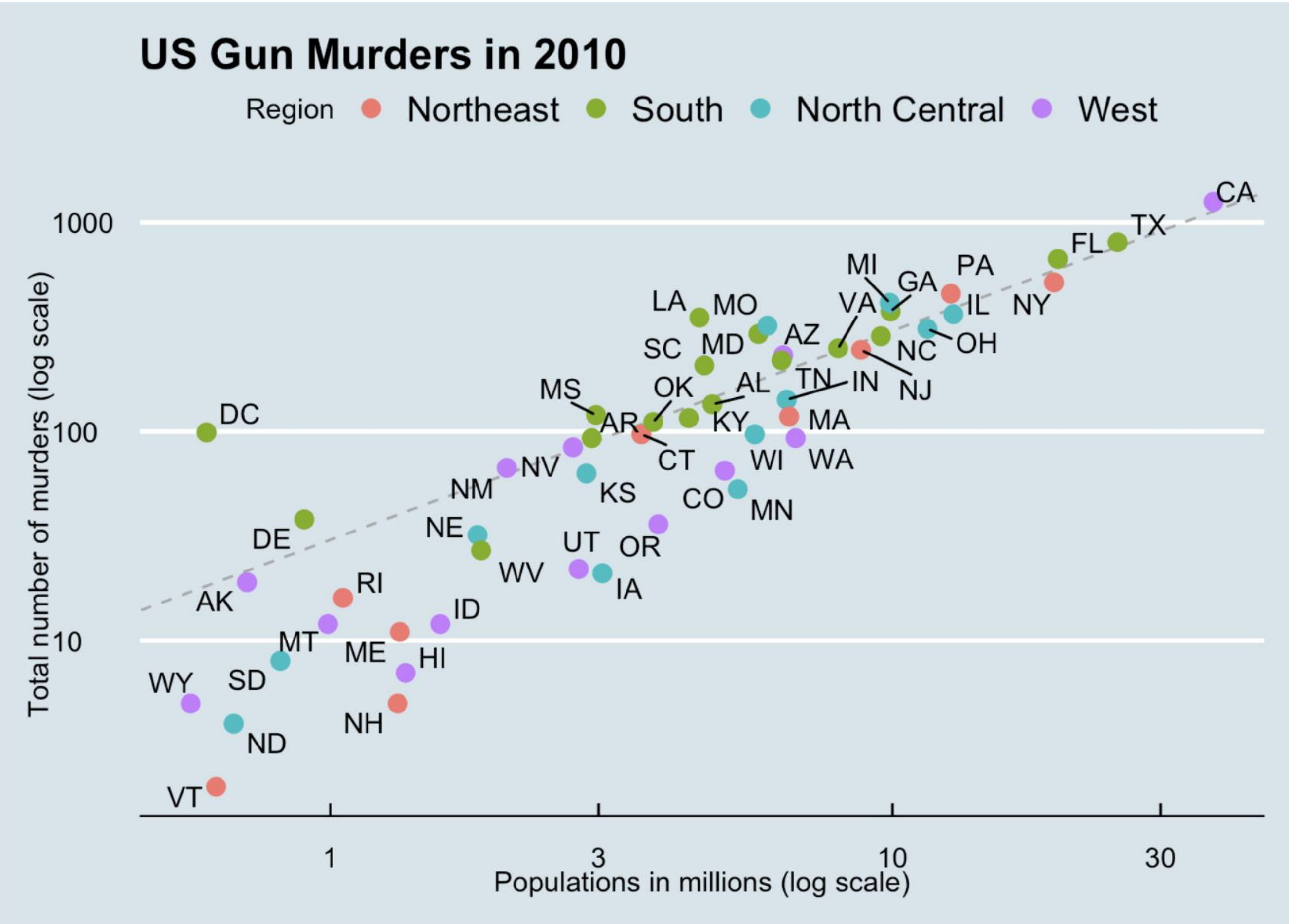
```
  Layer n
```



murders data example

```
library(tidyverse)
library(dslabs)
data(murders)
head(murders)
```

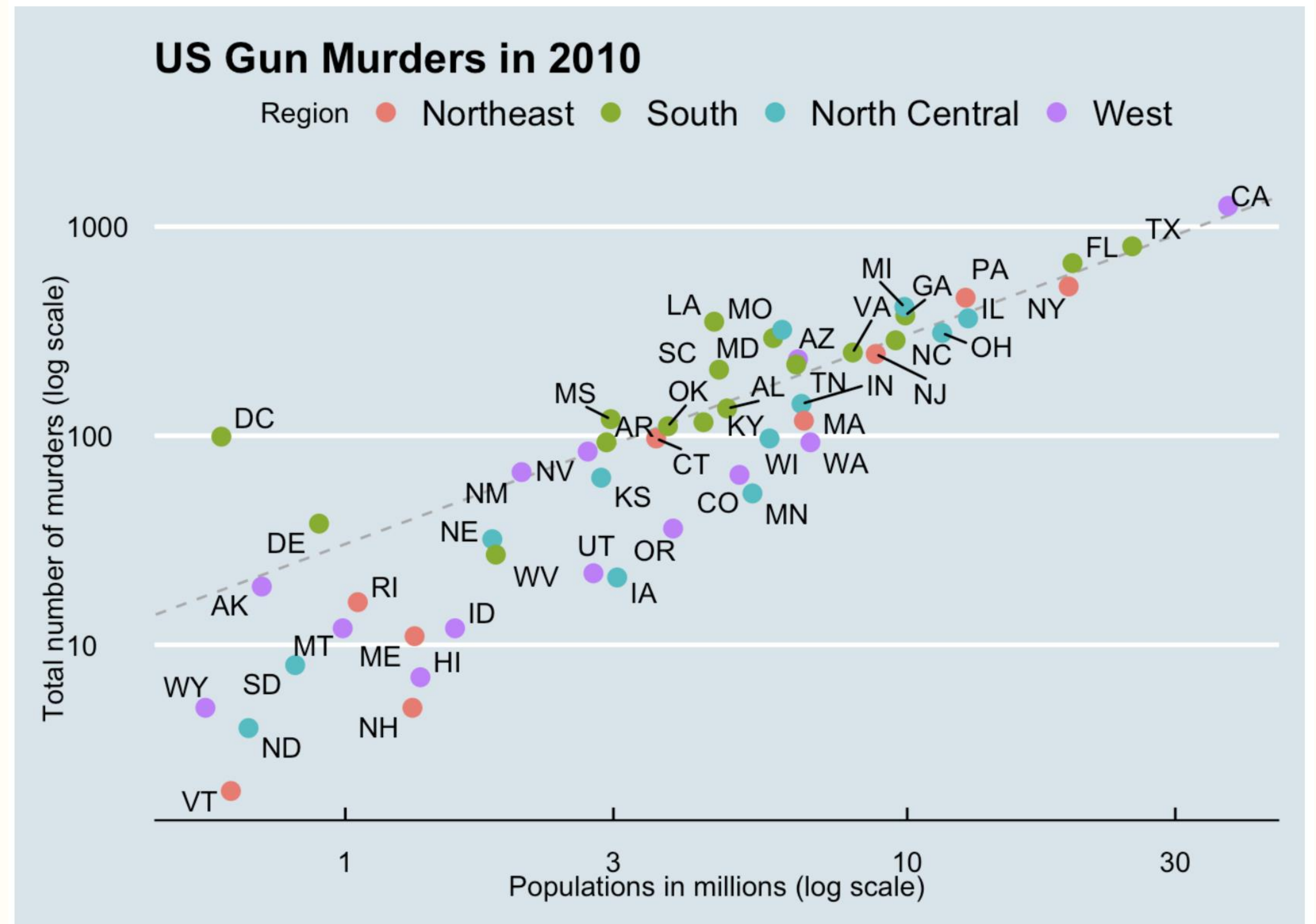
##	state	abb	region	population	total
## 1	Alabama	AL	South	4779736	135
## 2	Alaska	AK	West	710231	19
## 3	Arizona	AZ	West	6392017	232
## 4	Arkansas	AR	South	2915918	93
## 5	California	CA	West	37253956	1257
## 6	Colorado	CO	West	5029196	65



murders data example

List of components

- Population in millions on x-axis (log scale)
- Total murders on y-axis (log scale)
- Points color based on region
- Points labeled with state abbreviation
- Title and legend at the top
- Theme – background color and organization of the plot
- “Pretty” axis labels, title, and legend



ggplot data type

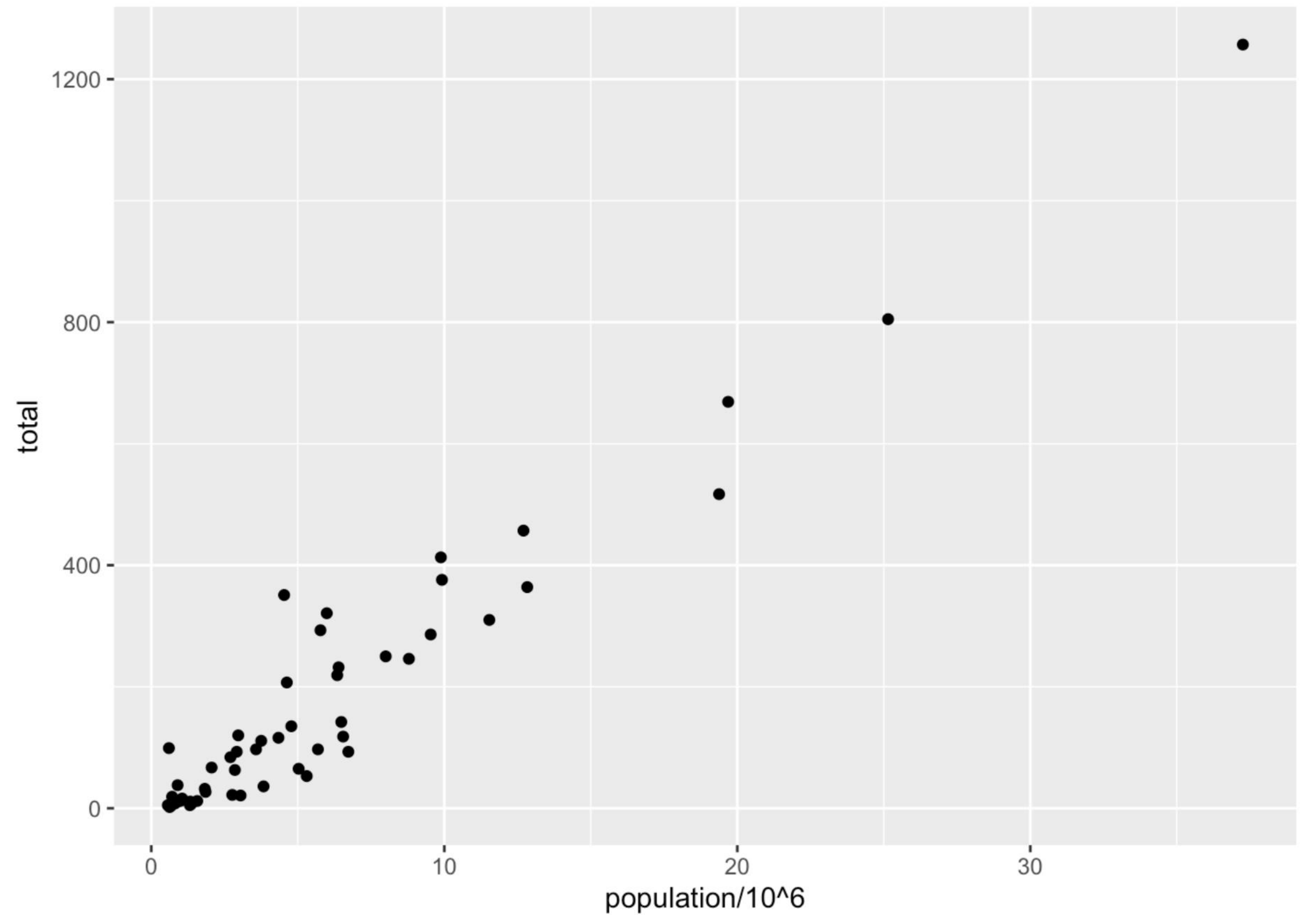
```
murders %>% ggplot()
```

```
p <- ggplot(data = murders)
class(p)
```

```
## [1] "gg"      "ggplot"
```

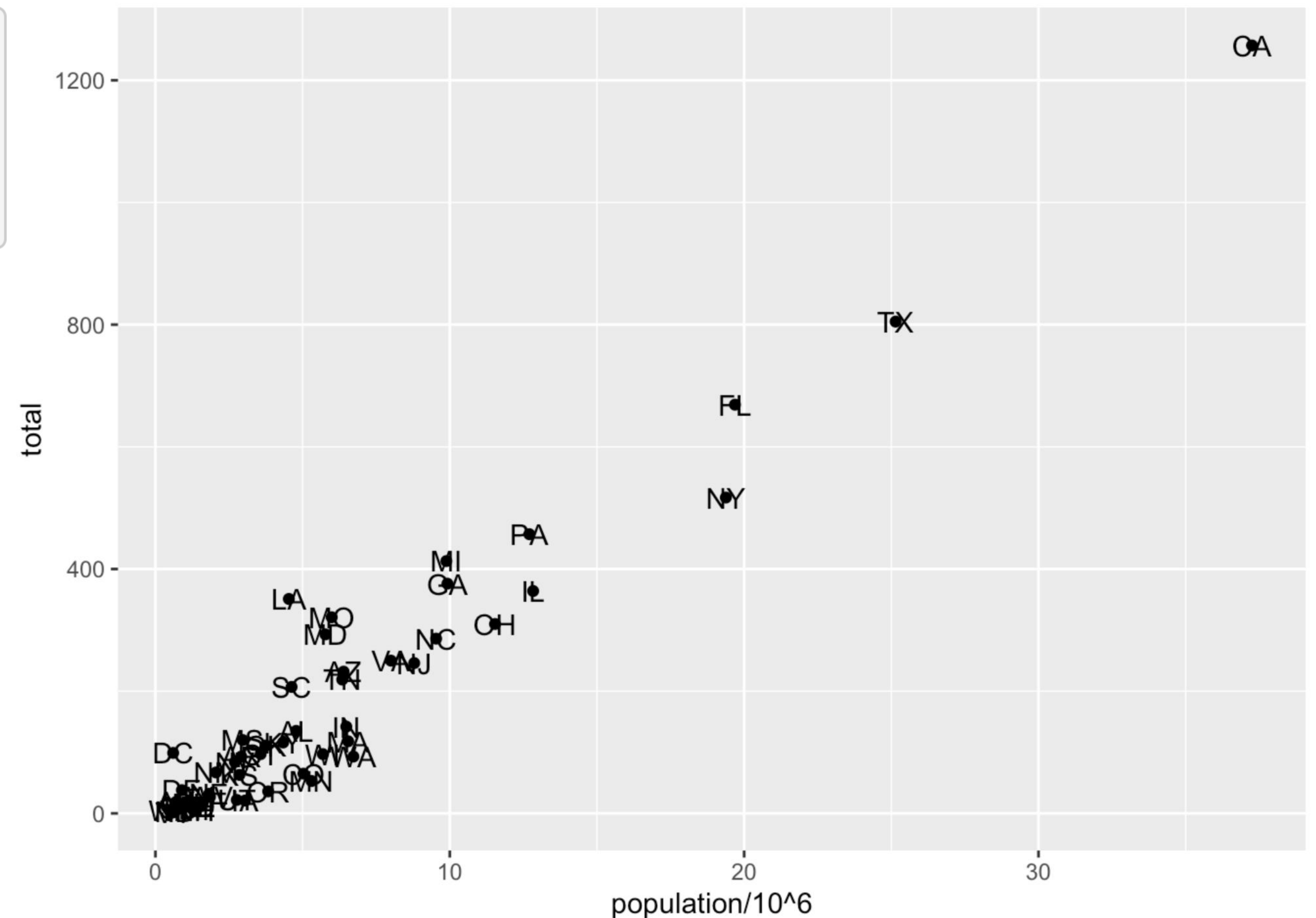
Scatterplot geometry and aesthetic mapping

```
murders %>%  
  ggplot() +  
  geom_point(aes(x = population/10^6, y = total))
```



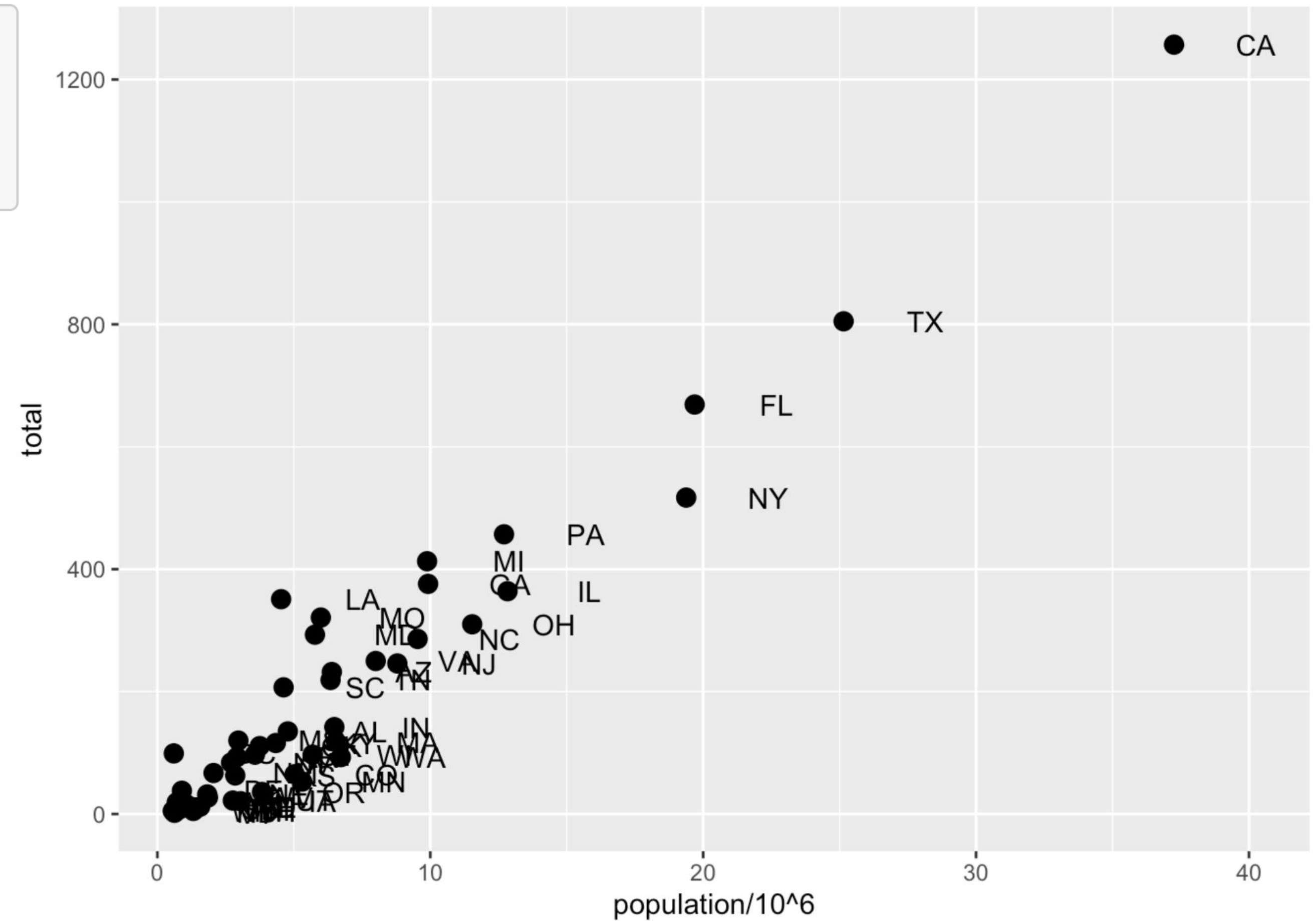
Adding labels

```
murders %>%  
  ggplot() +  
  geom_point(aes(population/10^6, total)) +  
  geom_text(aes(population/10^6, total, label = abb))
```



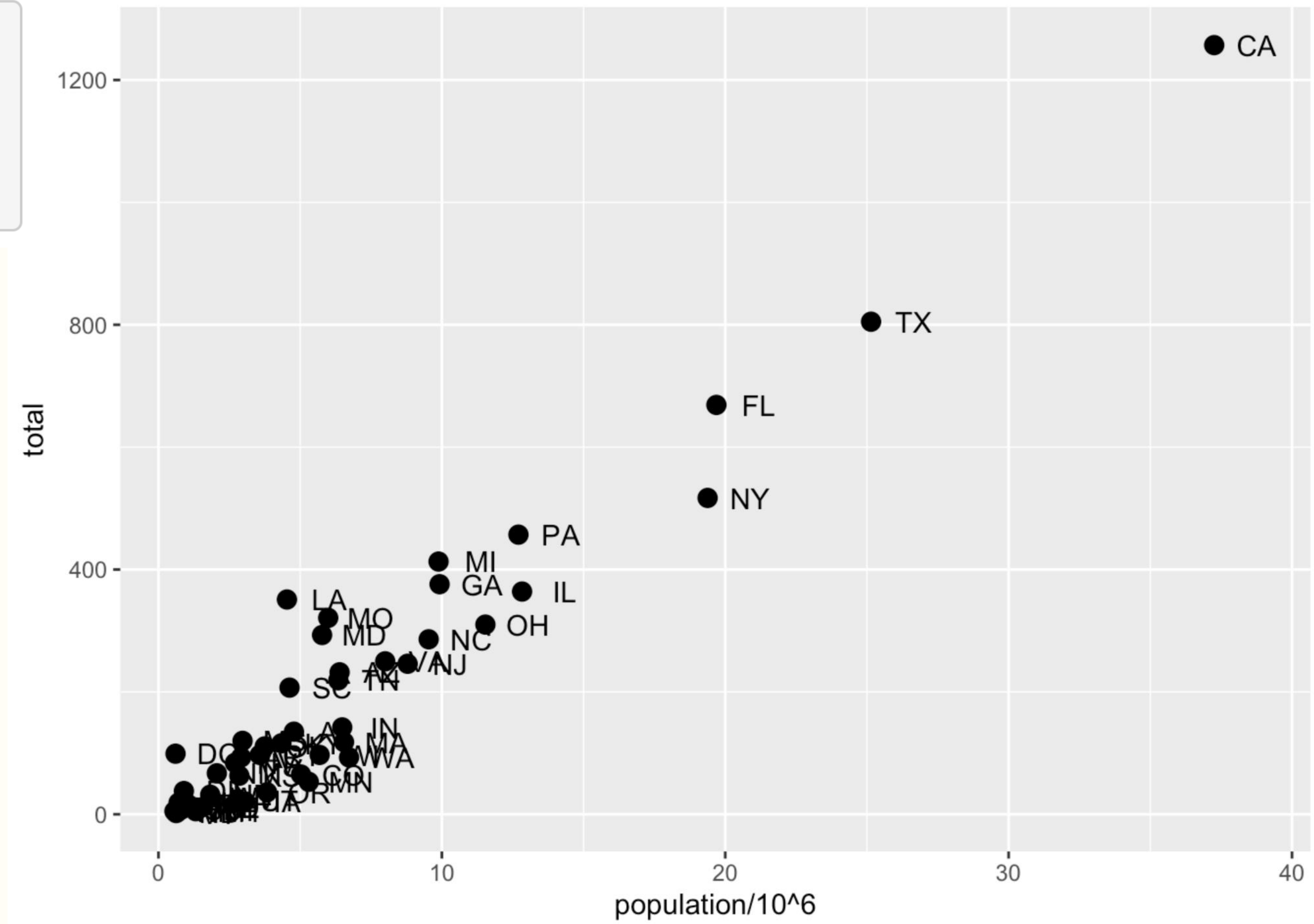
Adding labels

```
murders %>%  
  ggplot() +  
  geom_point(aes(population/10^6, total), size = 3) +  
  geom_text(aes(population/10^6, total, label = abb), nudge_x = 3)
```



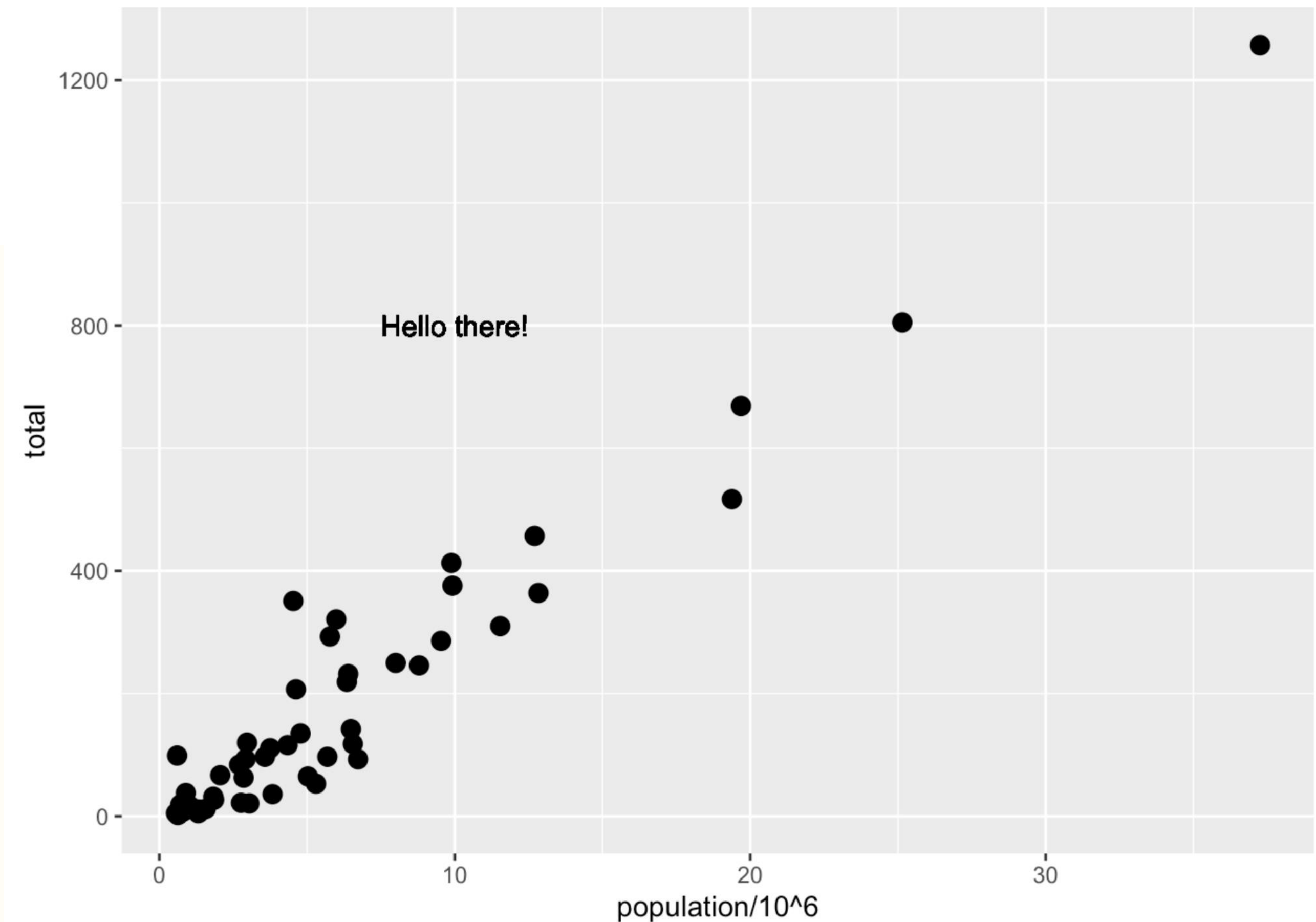
Global aesthetic mappings

```
murders %>%  
  ggplot(aes(x = population/10^6, y = total, label = abb)) +  
  geom_point(size = 3) +  
  geom_text(nudge_x = 1.5)
```



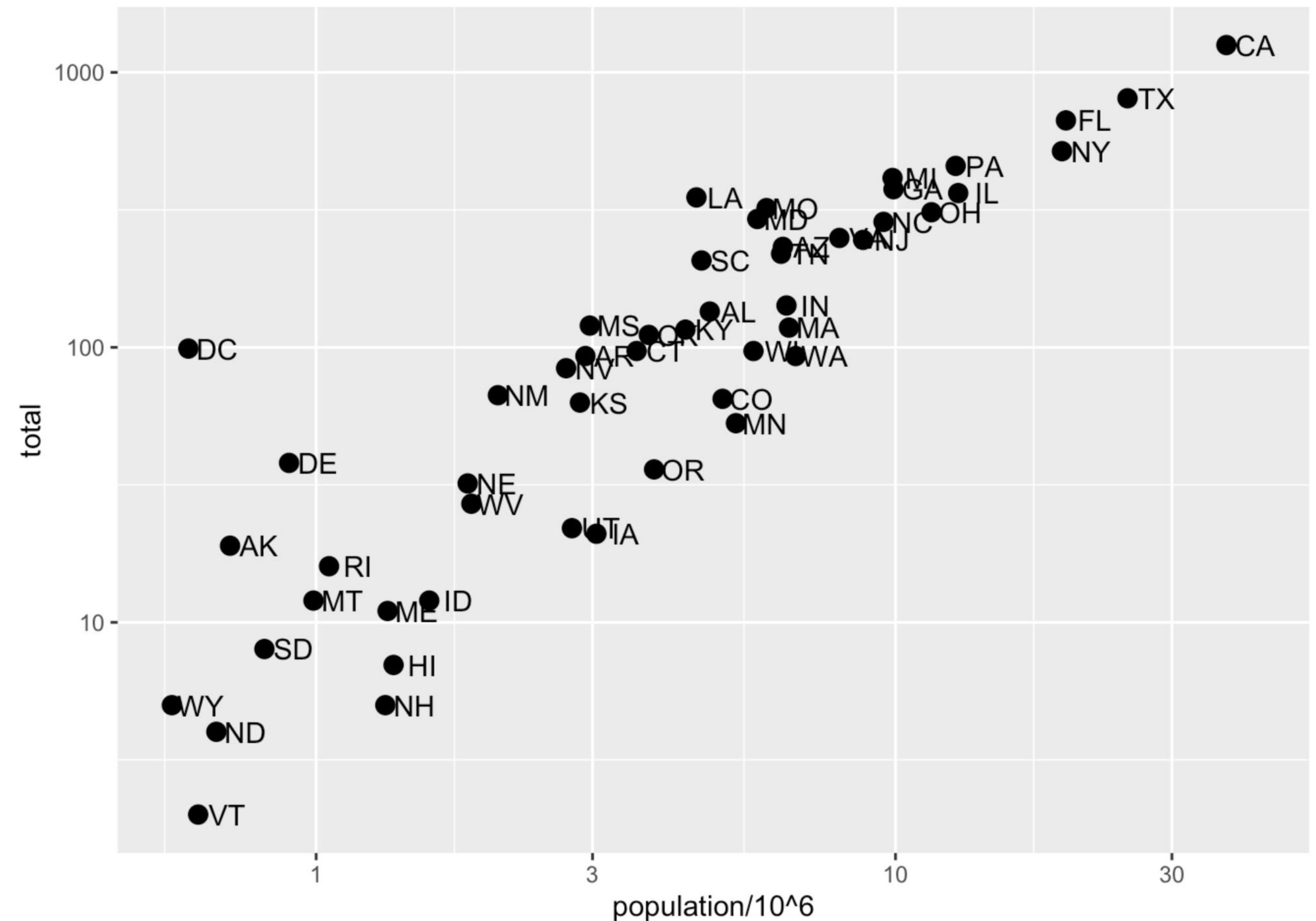
Local aesthetic mappings override global mappings

```
murders %>%  
  ggplot(aes(x = population/10^6, y = total, label = abb)) +  
  geom_point(size = 3) +  
  geom_text(aes(x = 10, y = 800, label = "Hello there!"))
```



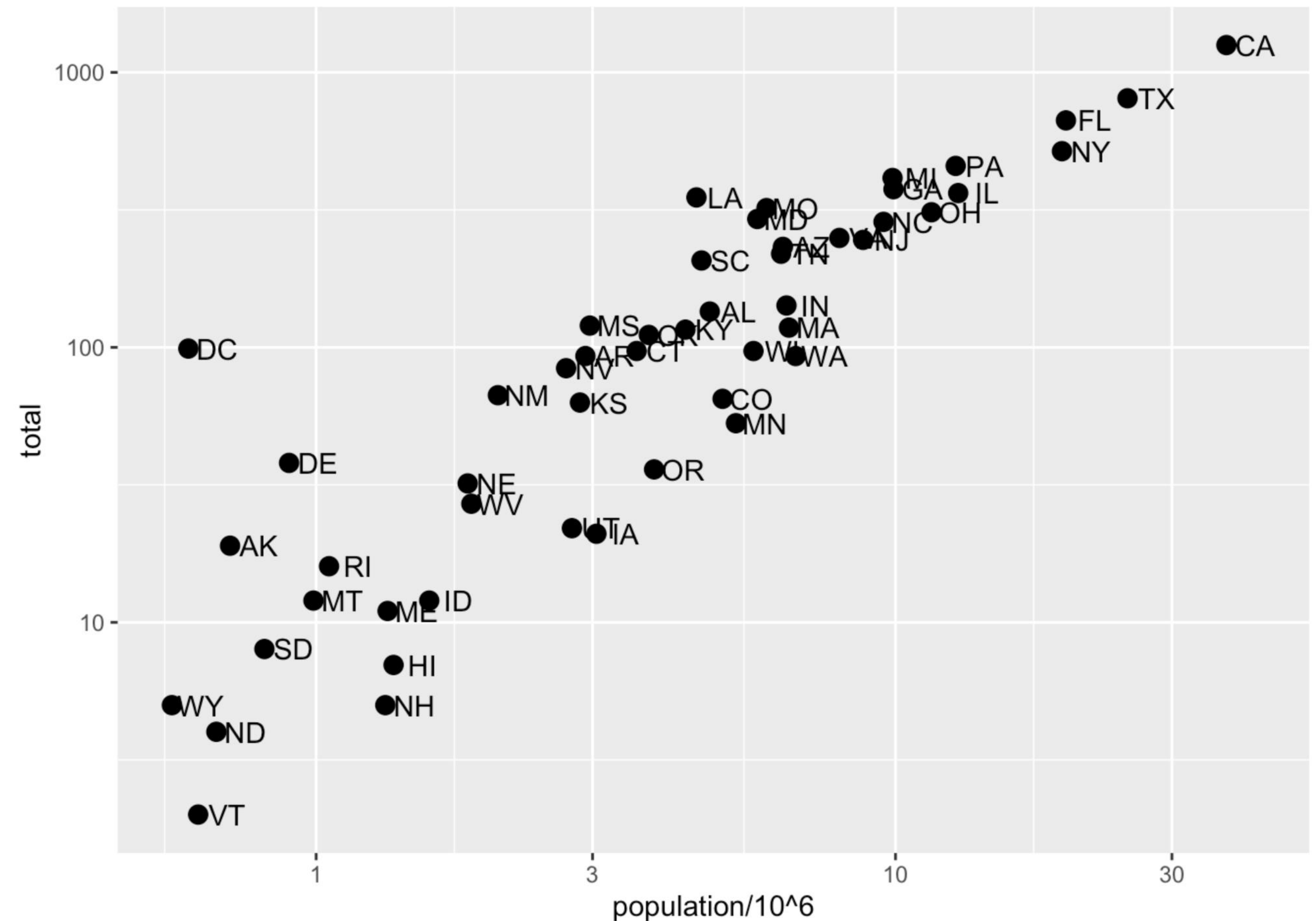
Scales

```
murders %>%  
  ggplot(aes(x = population/10^6, y = total, label = abb)) +  
  geom_point(size = 3) +  
  geom_text(nudge_x = 0.05) +  
  scale_x_continuous(trans = "log10") +  
  scale_y_continuous(trans = "log10")
```



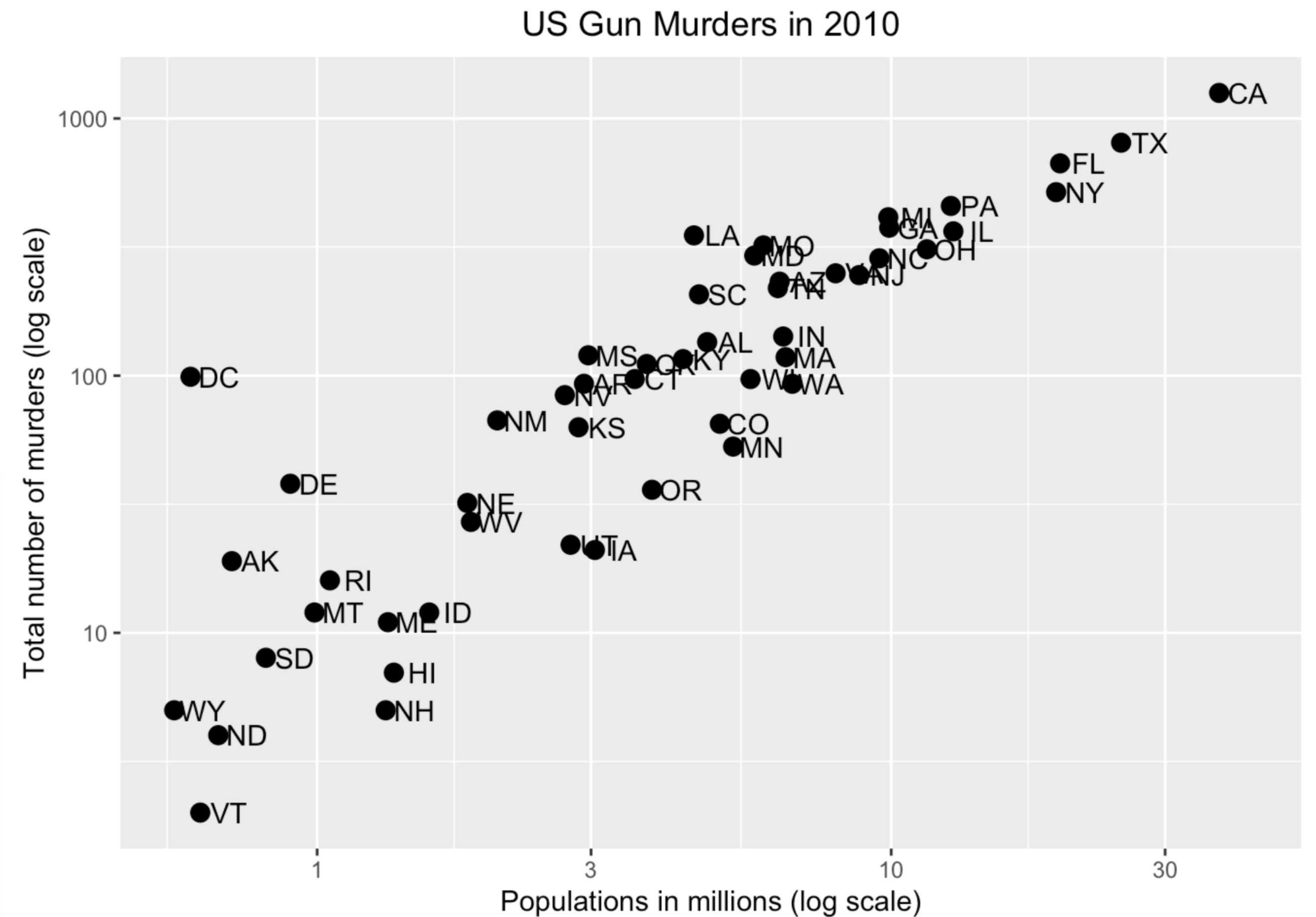
Scales

```
murders %>%  
  ggplot(aes(x = population/10^6, y = total, label = abb)) +  
  geom_point(size = 3) +  
  geom_text(nudge_x = 0.05) +  
  scale_x_log10() +  
  scale_y_log10()
```

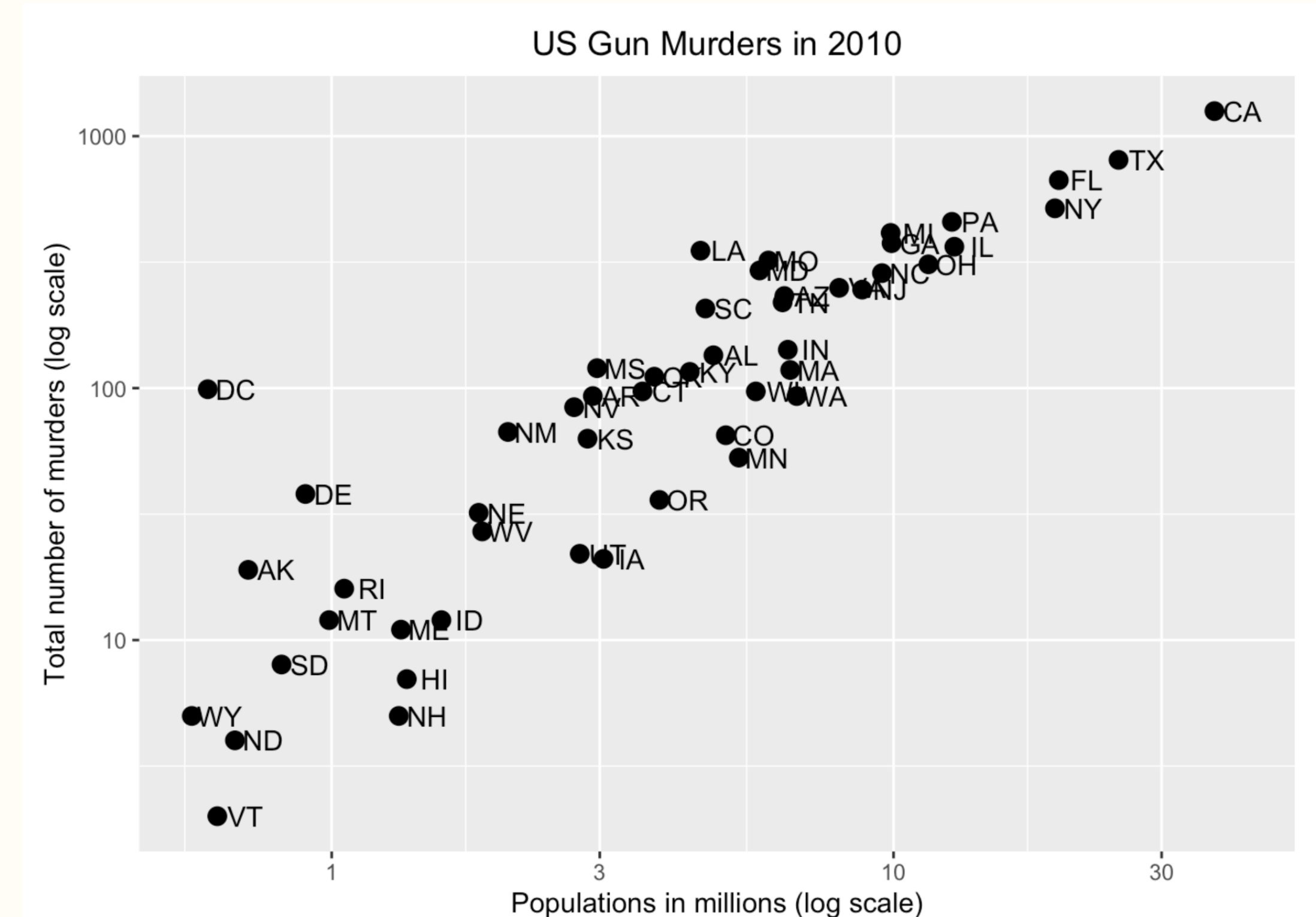
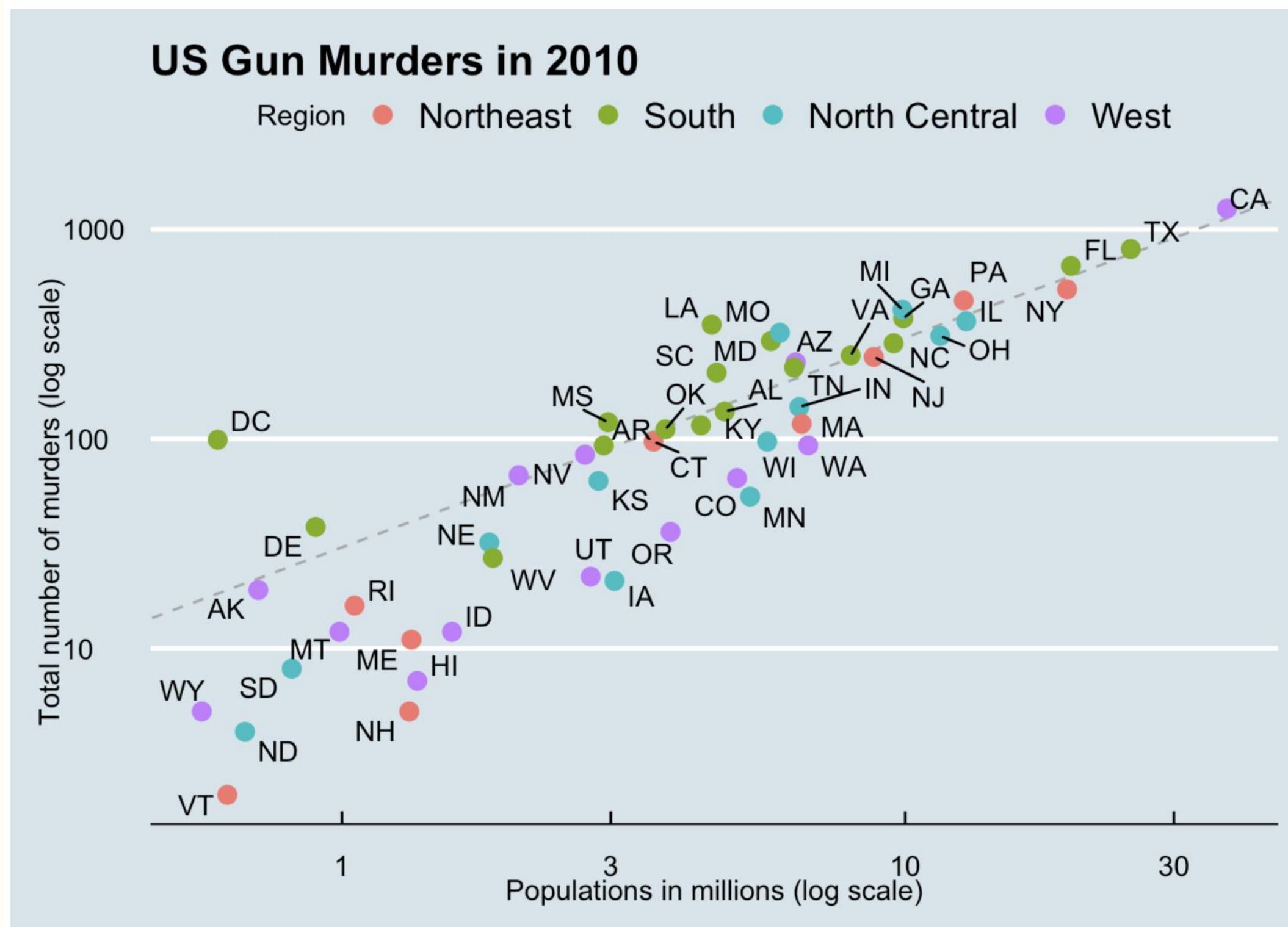


Axis labels and titles

```
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_point(size = 3) +
  geom_text(nudge_x = 0.05) +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  theme(plot.title = element_text(hjust = 0.5))
```



Progress check



Summary

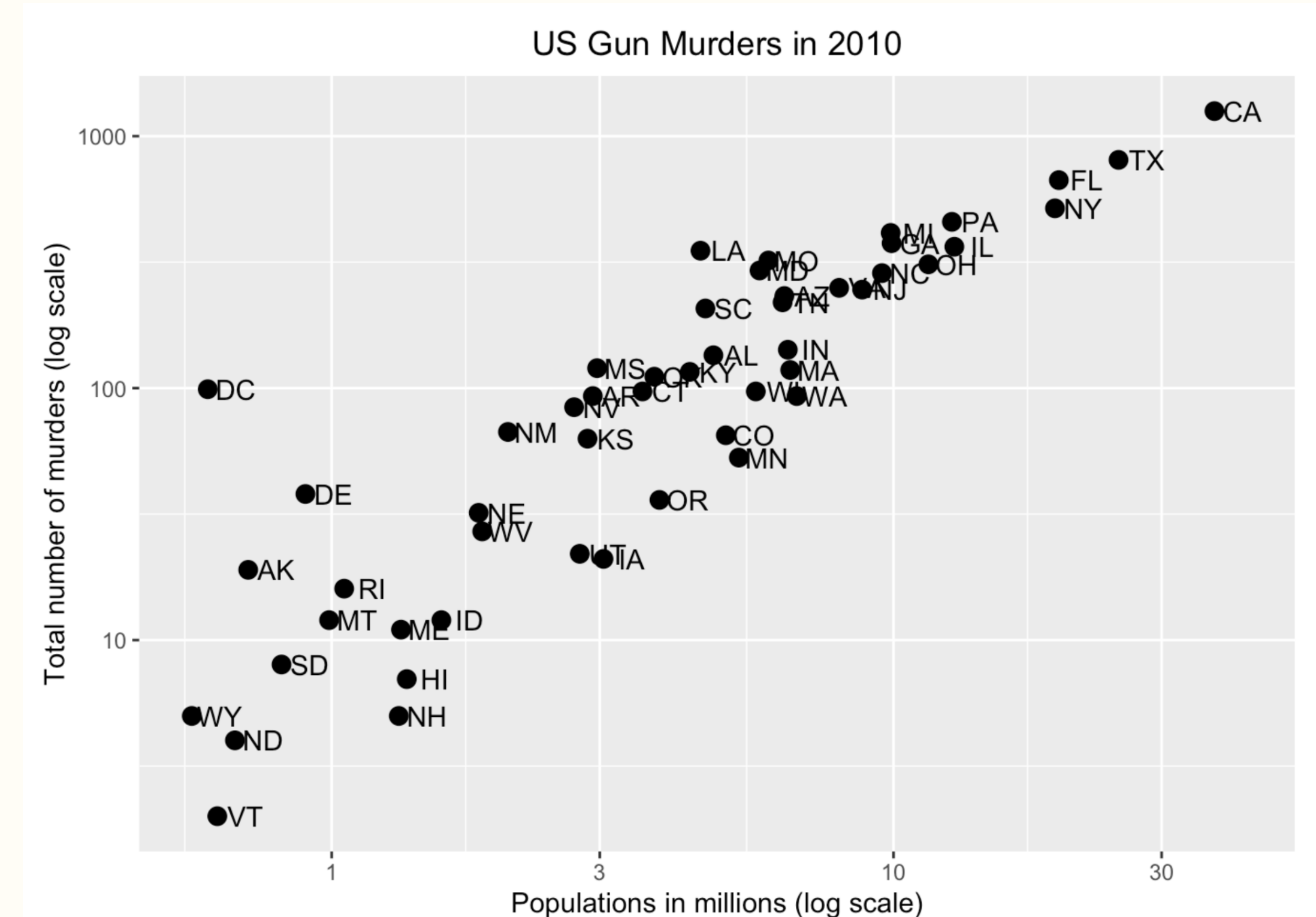
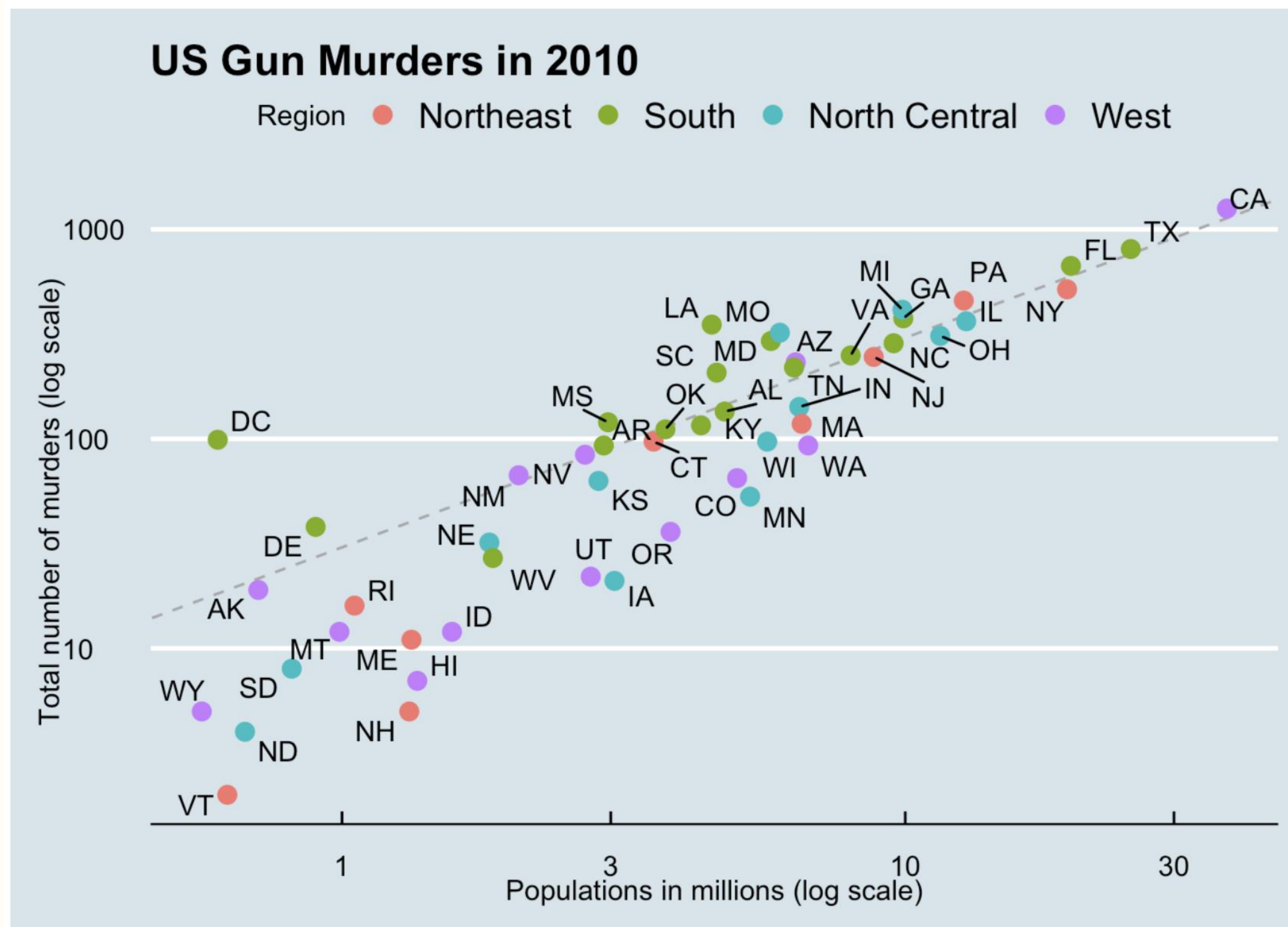
- The **ggplot2** package is a tool to create high-quality, effective visualizations in a concise manner
- The anatomy of a **ggplot** includes data, geometry, aesthetic mapping and layers of customizations

Introduction to data visualization

Fine-tuning visualizations

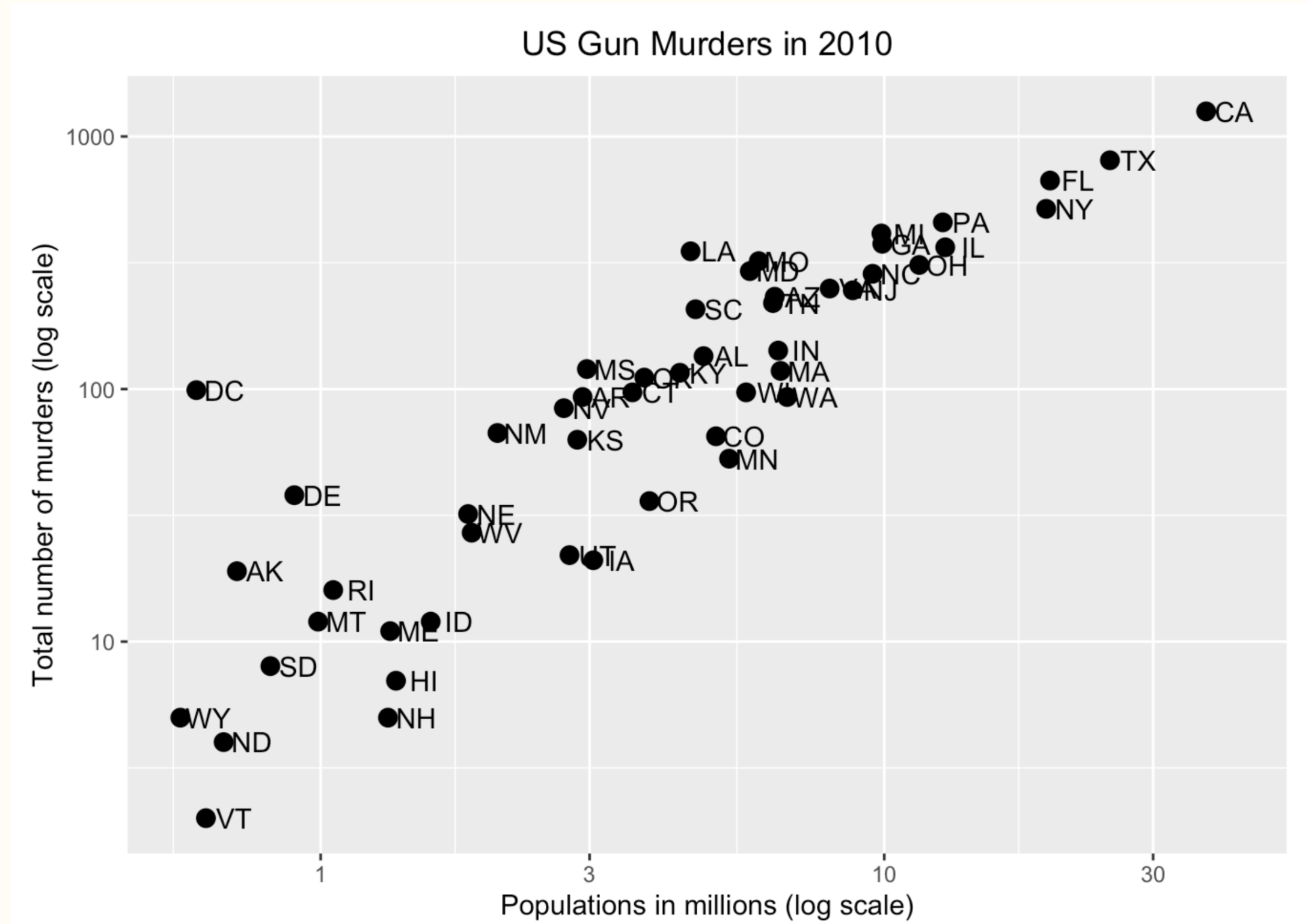
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Progress check



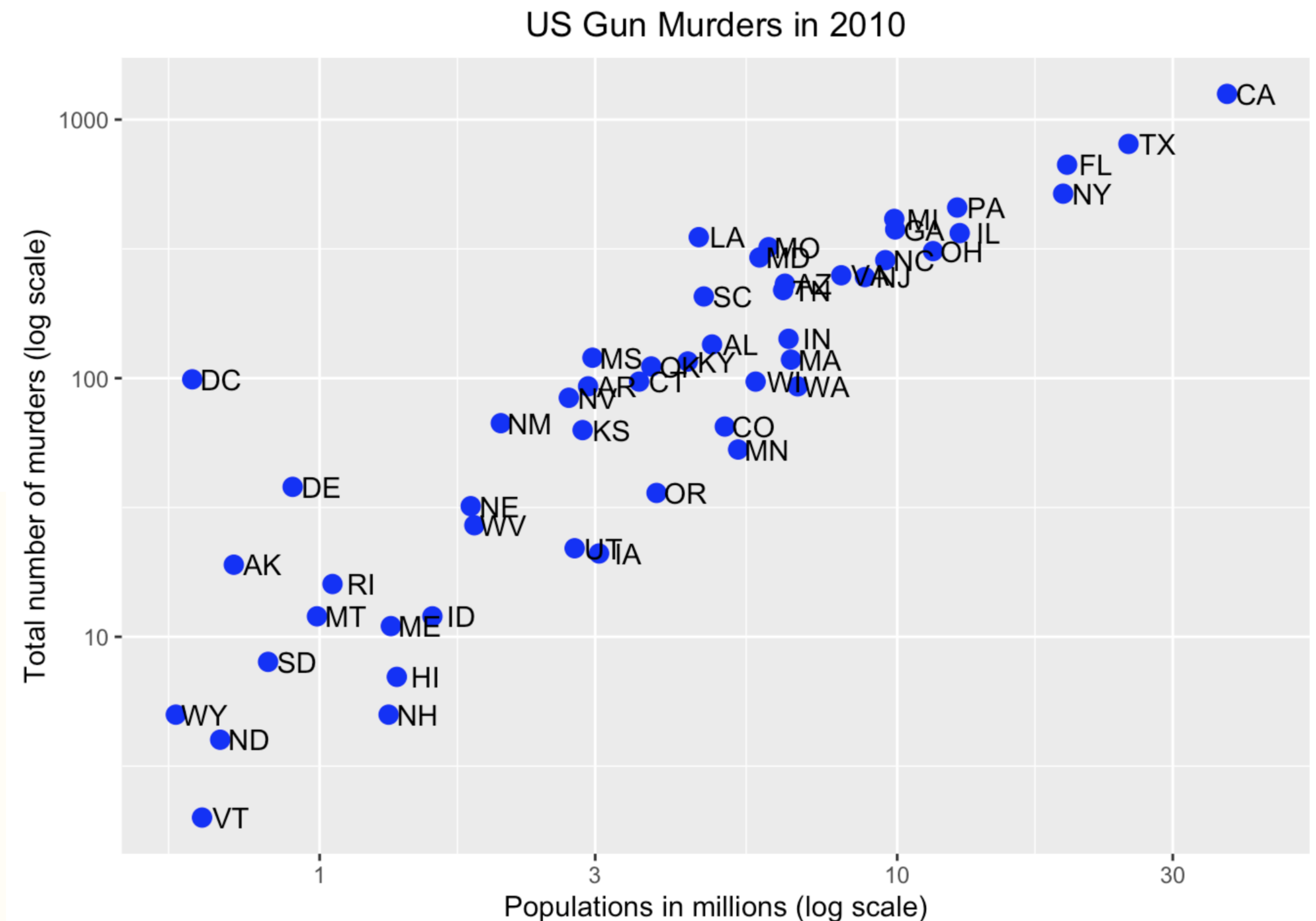
Progress check

```
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_point(size = 3) +
  geom_text(nudge_x = 0.05) +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  theme(plot.title = element_text(hjust = 0.5))
```



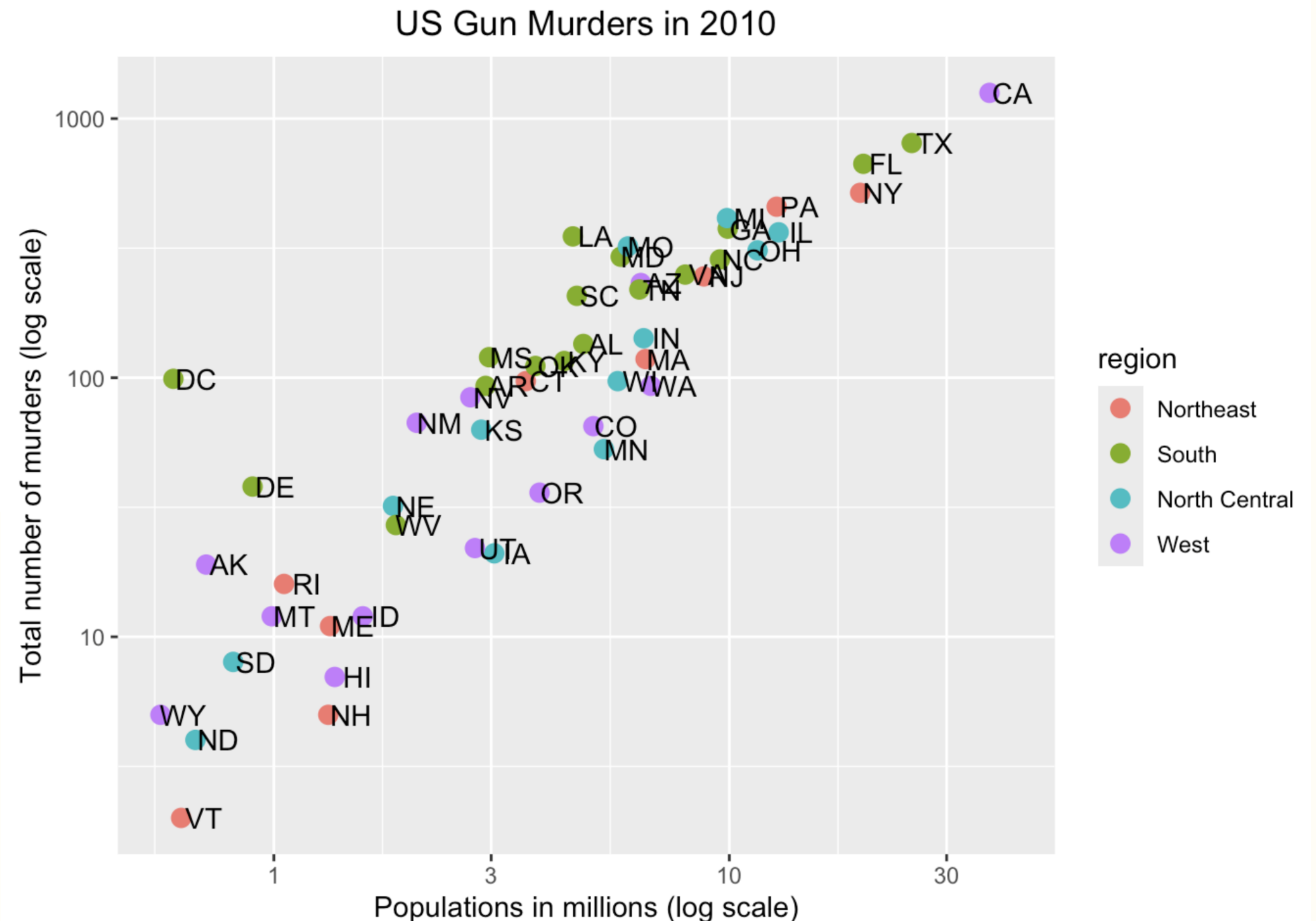
Categories and colors

```
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_point(size = 3, color = "blue") +
  geom_text(nudge_x = 0.05) +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  theme(plot.title = element_text(hjust = 0.5))
```



Categories and colors

```
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_point(aes(color = region), size = 3) +
  geom_text(nudge_x = 0.05) +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  theme(plot.title = element_text(hjust = 0.5))
```

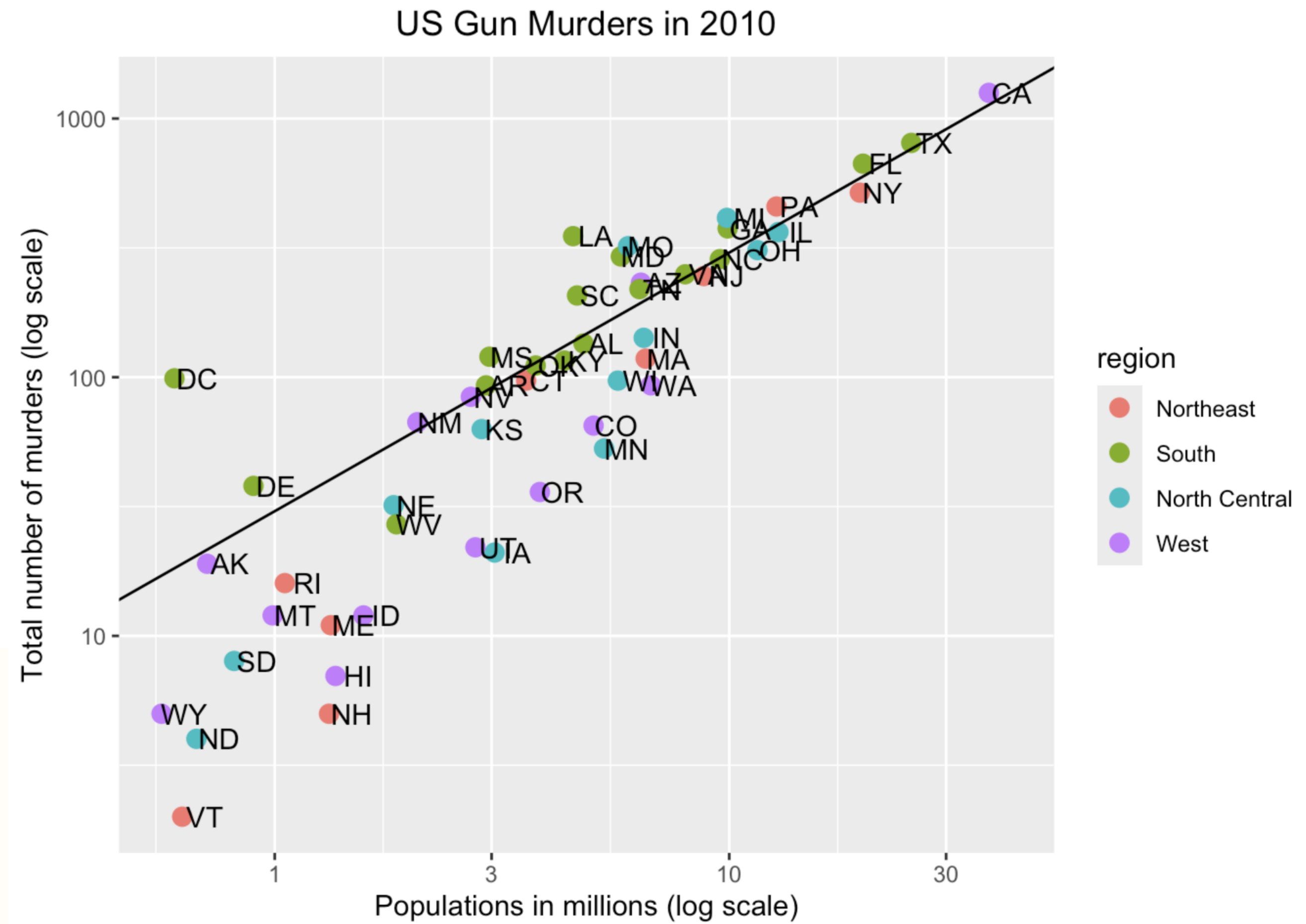


Adding a line

```
r <- murders %>%  
  summarize(murder_rate = sum(total) / sum(population) * 10^6) %>% .$.murder_rate  
r
```

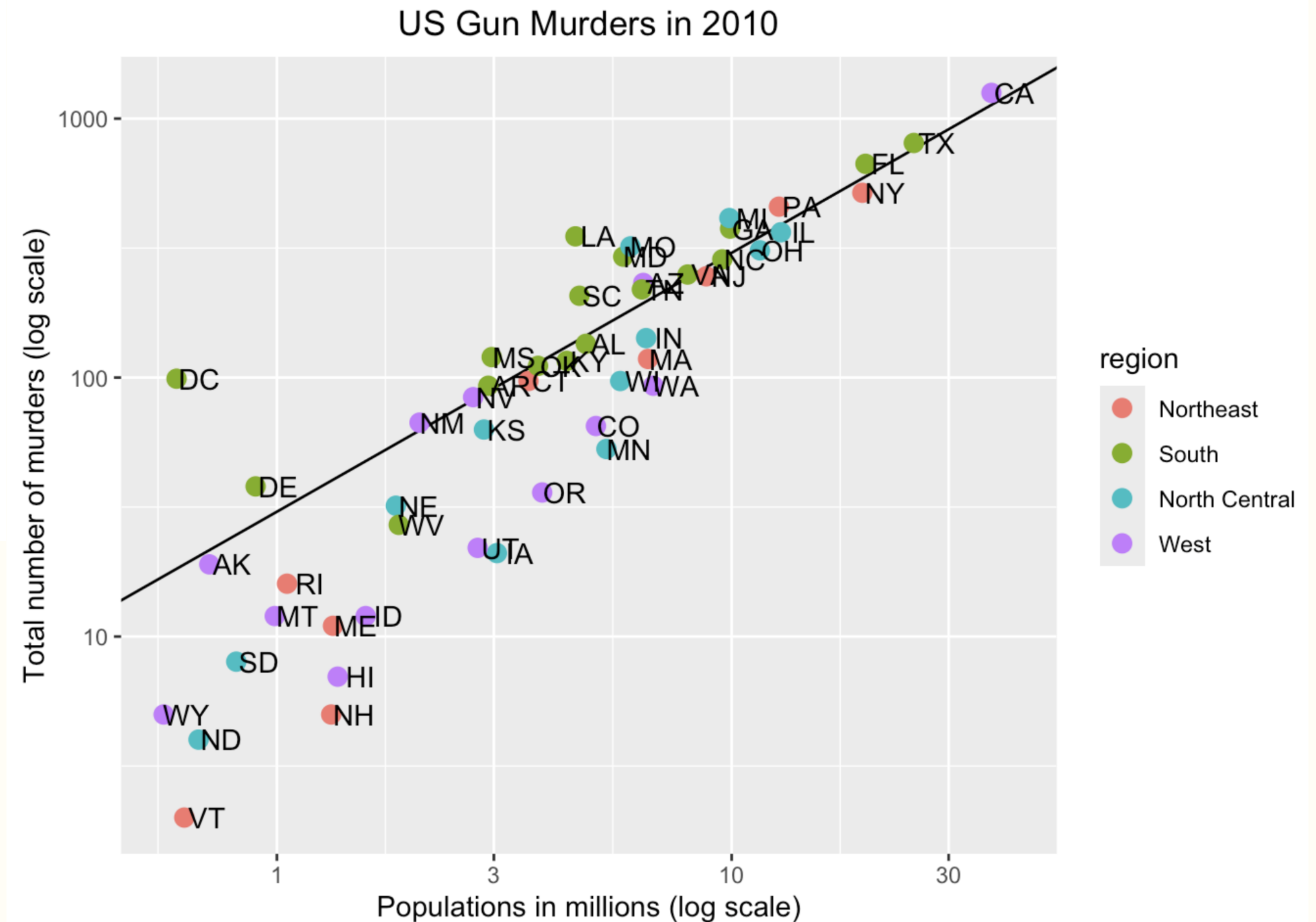
```
## [1] 30.34555
```

```
murders %>%  
  ggplot(aes(x = population/10^6, y = total, label = abb)) +  
  geom_point(aes(color = region), size = 3) +  
  geom_abline(intercept = log10(r)) +  
  geom_text(nudge_x = 0.05) +  
  scale_x_log10() +  
  scale_y_log10() +  
  xlab("Populations in millions (log scale)") +  
  ylab("Total number of murders (log scale)") +  
  ggtitle("US Gun Murders in 2010") +  
  theme(plot.title = element_text(hjust = 0.5))
```



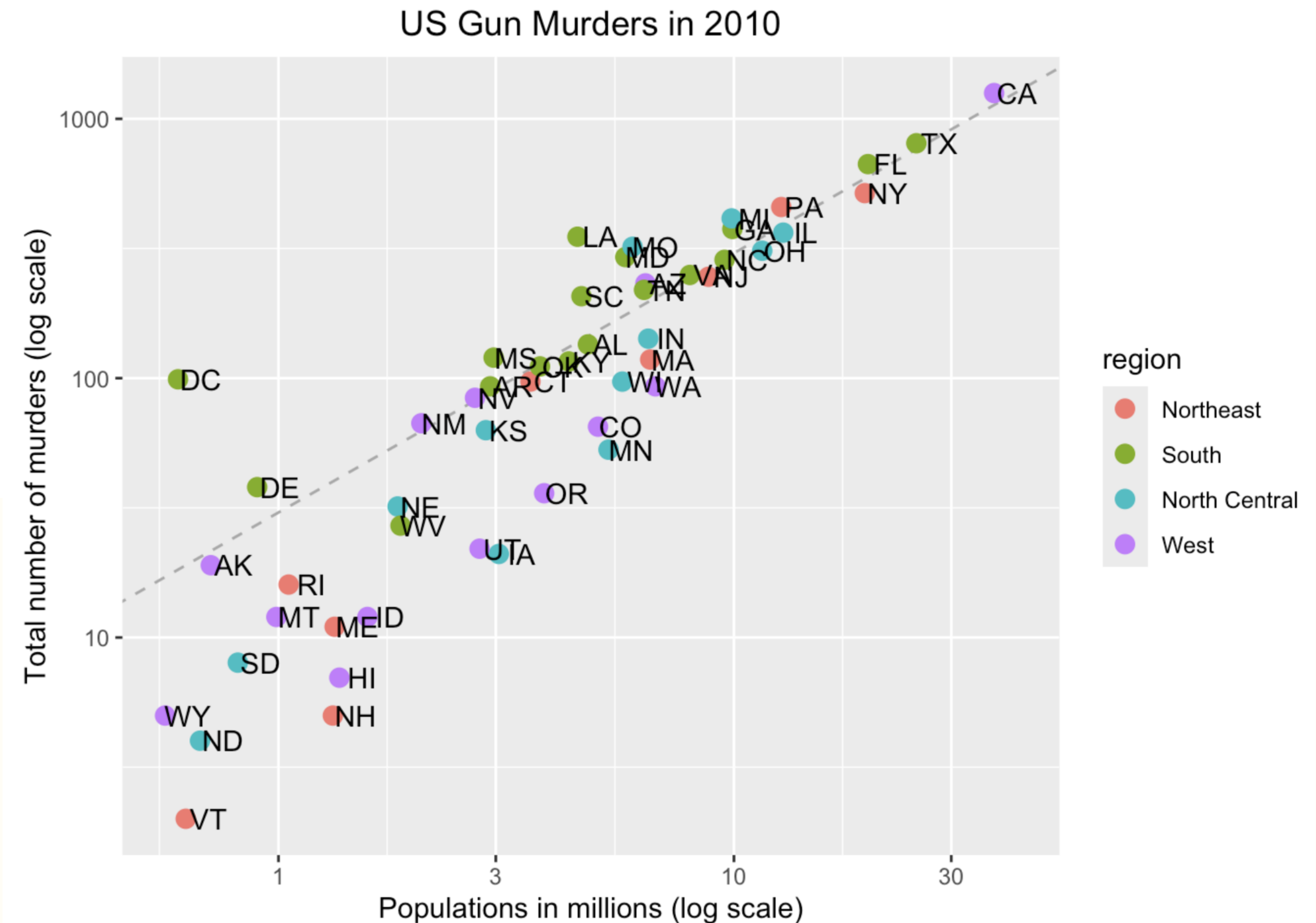
Adding a line

```
murders %>%  
  ggplot(aes(x = population/10^6, y = total, label = abb)) +  
  geom_abline(intercept = log10(r)) +  
  geom_point(aes(color = region), size = 3) +  
  geom_text(nudge_x = 0.05) +  
  scale_x_log10() +  
  scale_y_log10() +  
  xlab("Populations in millions (log scale)") +  
  ylab("Total number of murders (log scale)") +  
  ggtitle("US Gun Murders in 2010") +  
  theme(plot.title = element_text(hjust = 0.5))
```



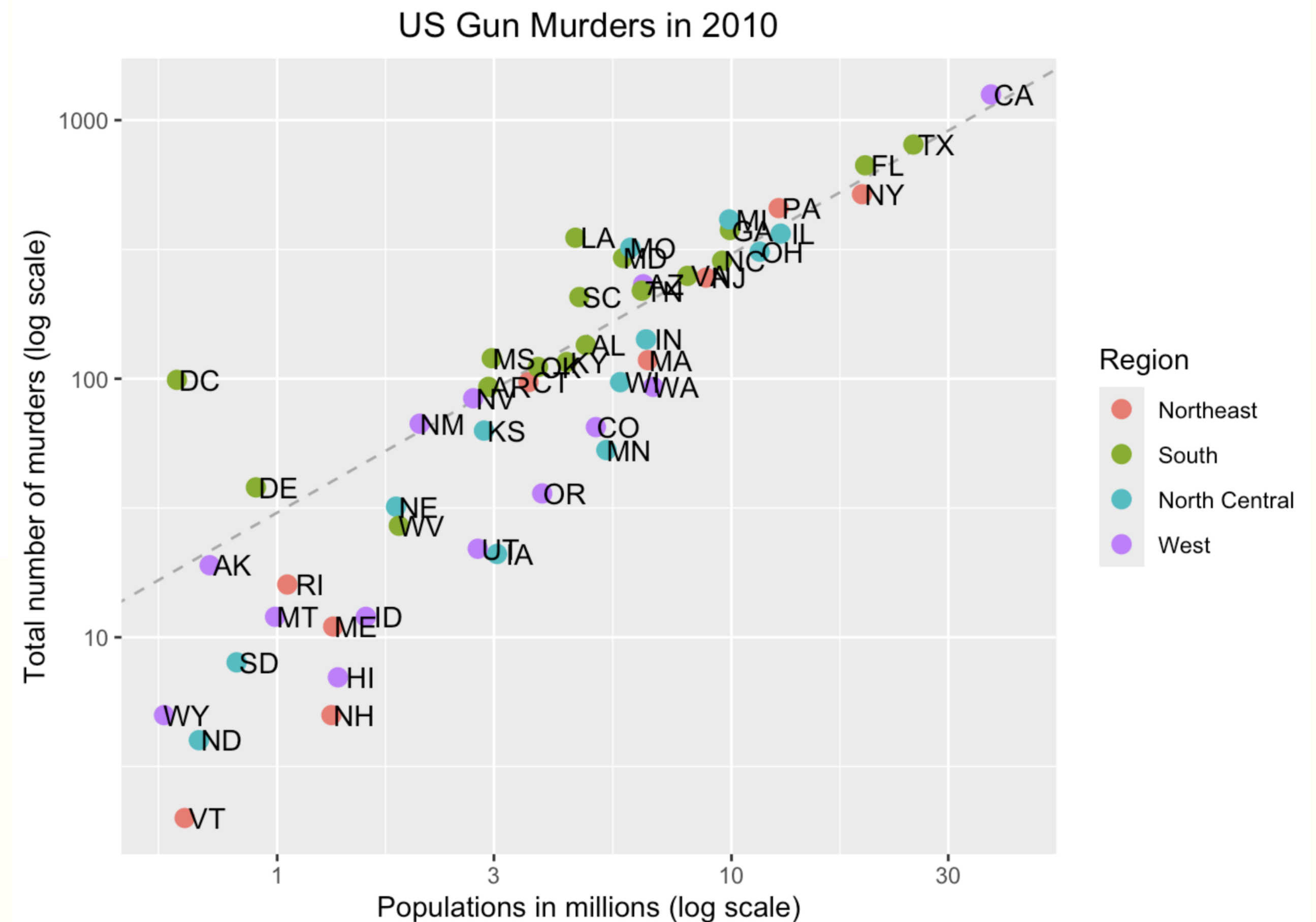
Adding a line

```
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_abline(intercept = log10(r), lty = 2, color = "darkgrey") +
  geom_point(aes(color = region), size = 3) +
  geom_text(nudge_x = 0.05) +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  theme(plot.title = element_text(hjust = 0.5))
```



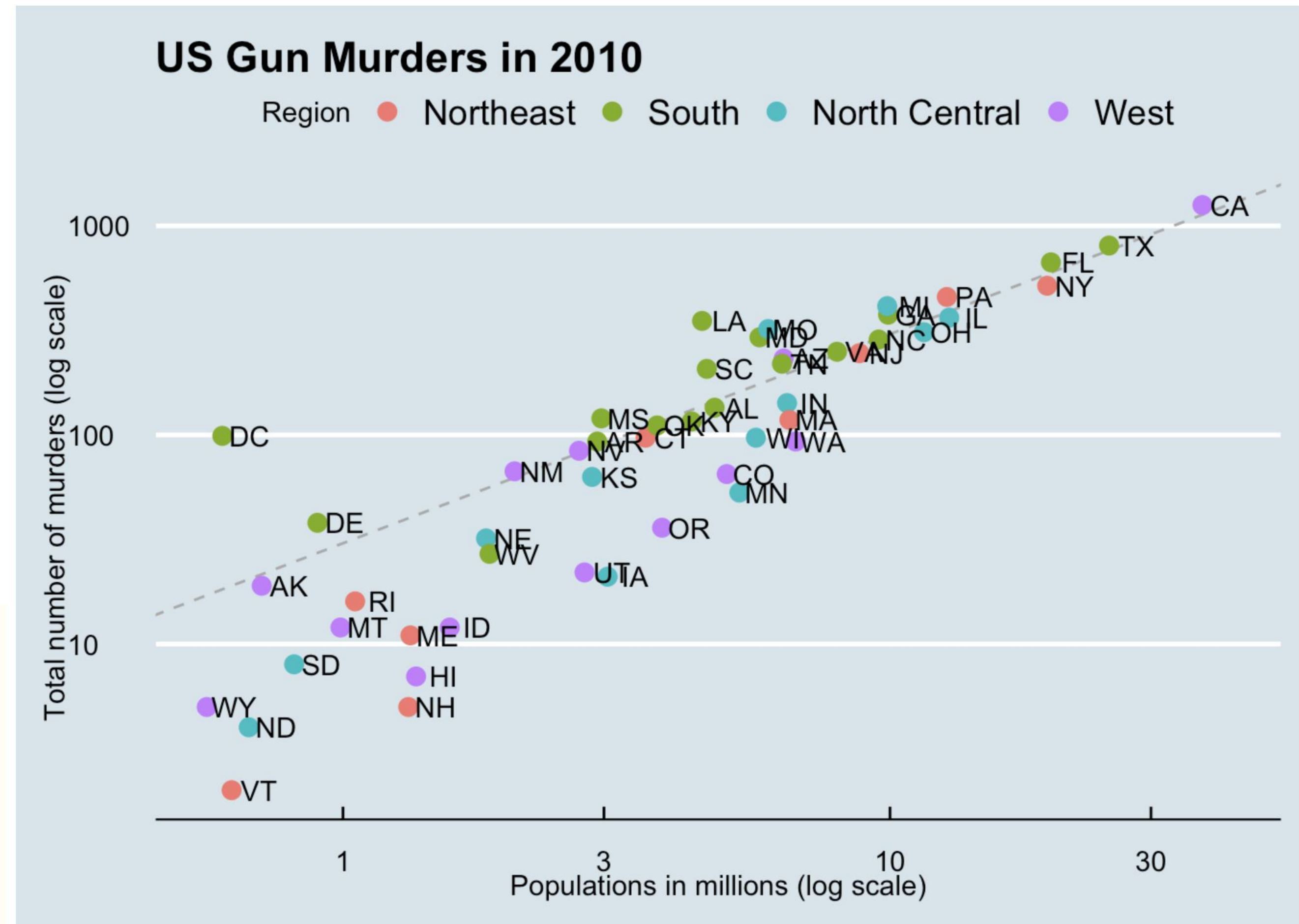
Legend title

```
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_abline(intercept = log10(r), lty = 2, color = "darkgrey") +
  geom_point(aes(color = region), size = 3) +
  geom_text(nudge_x = 0.05) +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  theme(plot.title = element_text(hjust = 0.5)) +
  scale_color_discrete(name = "Region")
```



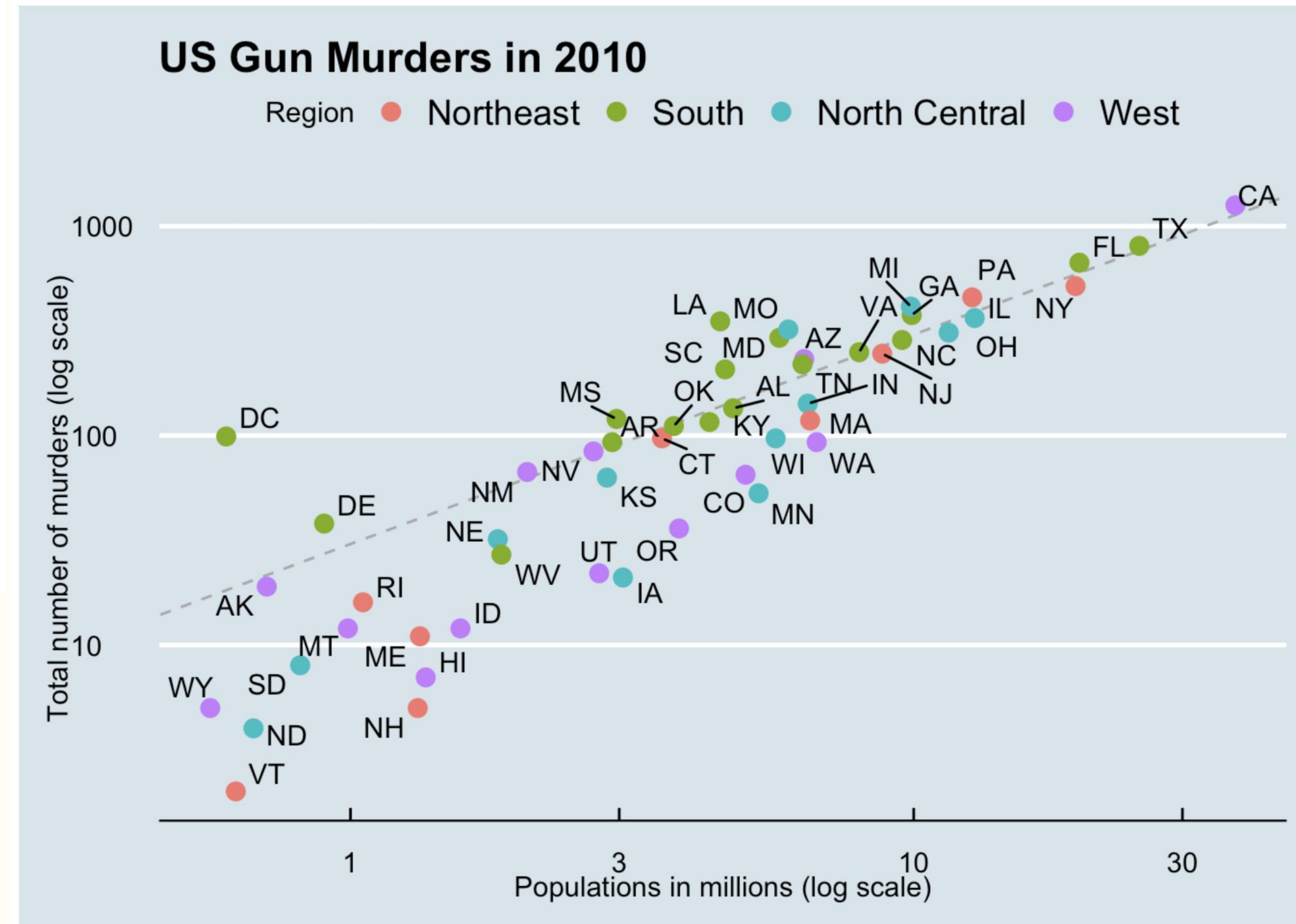
ggthemes

```
library(ggthemes)
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_abline(intercept = log10(r), lty = 2, color = "darkgrey") +
  geom_point(aes(color = region), size = 3) +
  geom_text(nudge_x = 0.05) +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  theme(plot.title = element_text(hjust = 0.5)) +
  scale_color_discrete(name = "Region") +
  theme_economist()
```



ggrepel

```
library(ggrepel)
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_abline(intercept = log10(r), lty = 2, color = "darkgrey") +
  geom_point(aes(color = region), size = 3) +
  geom_text_repel() +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  scale_color_discrete(name = "Region") +
  theme_economist()
```



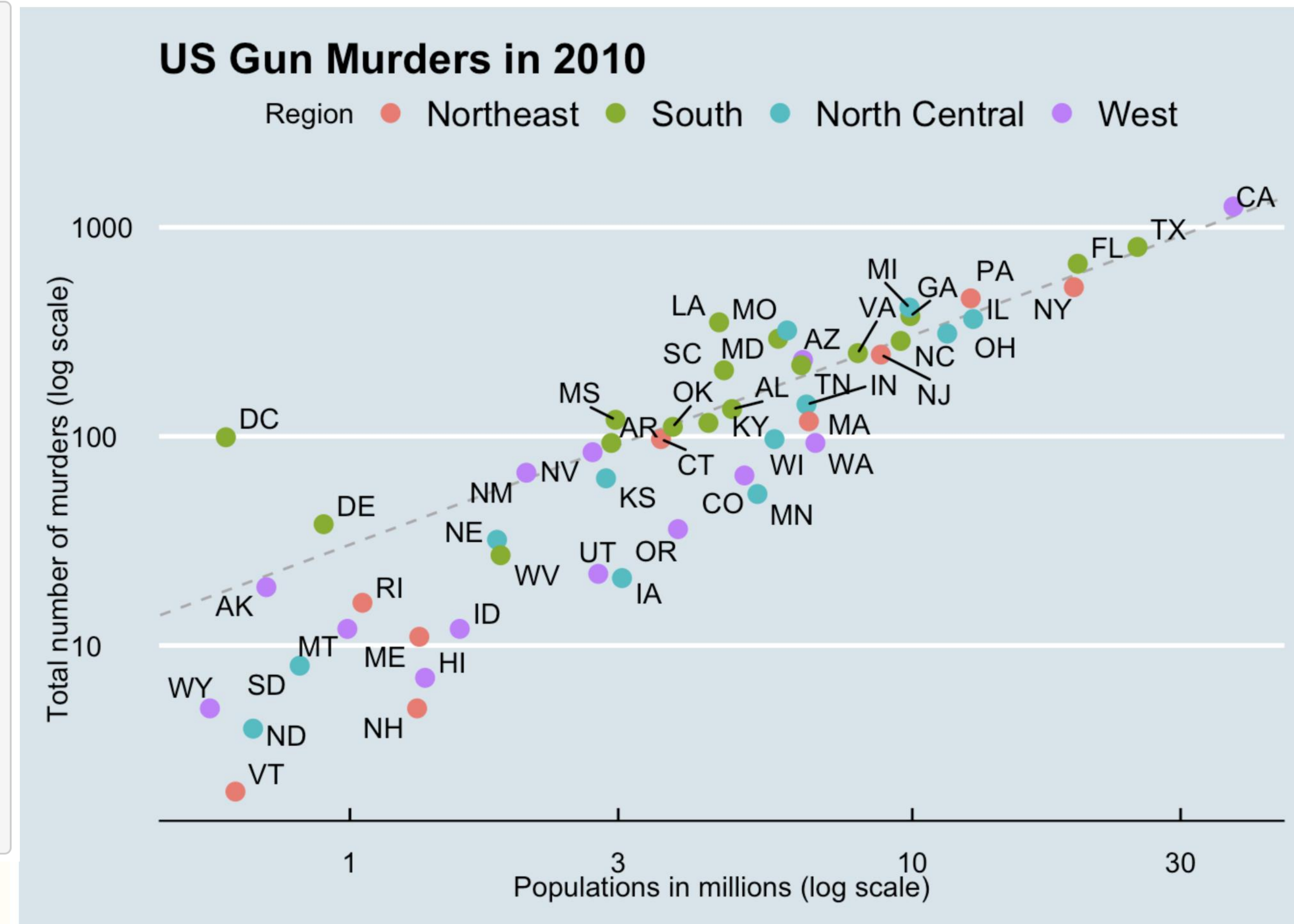
Putting it all together

```
library(tidyverse)
library(dslabs)
library(ggthemes)
library(ggrepel)

## Load the data
data(murders)

## First define the slope of the line
r <- murders %>%
  summarize(murder_rate= sum(total) / sum(population) * 10^6) %>% .$murder_rate

## Now make the plot
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_abline(intercept = log10(r), lty = 2, color = "darkgrey") +
  geom_point(aes(color = region), size = 3) +
  geom_text_repel() +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  scale_color_discrete(name = "Region") +
  theme_economist()
```



Summary

- Data visualizations created with `ggplot2` are highly customizable
- Plot appearance can be customized using themes, labels, colors, and scales